Linear Regression

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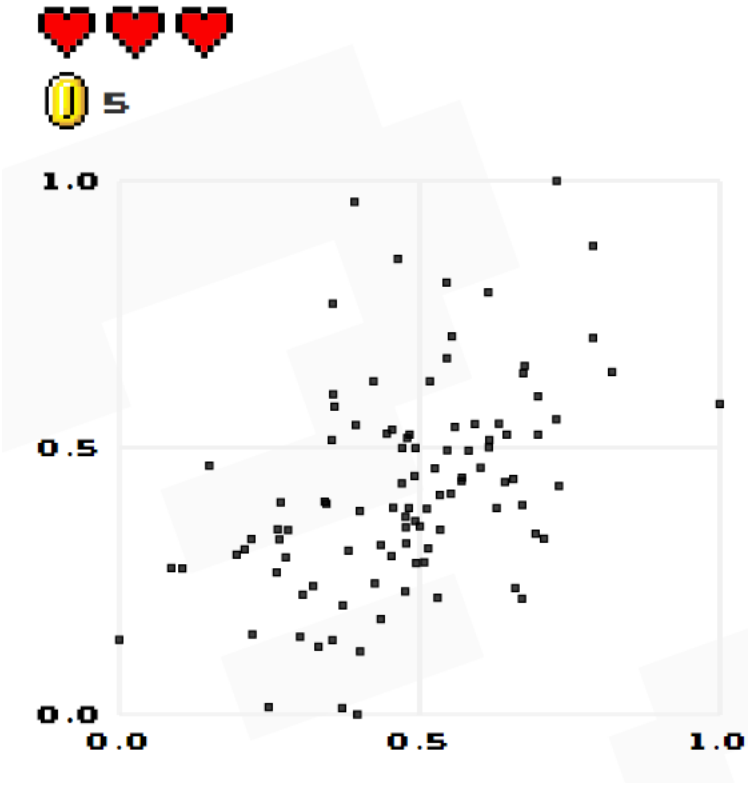
Dartmouth College. February 2026

Before we begin: Guess the correlation!

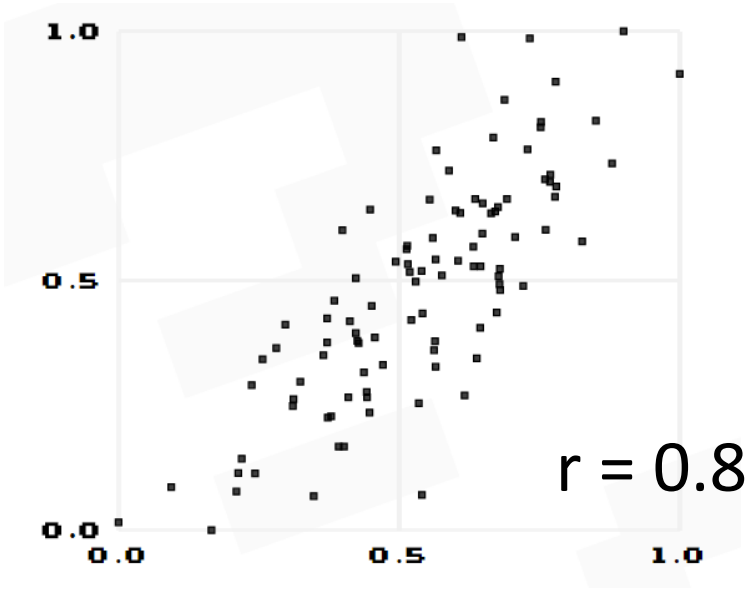


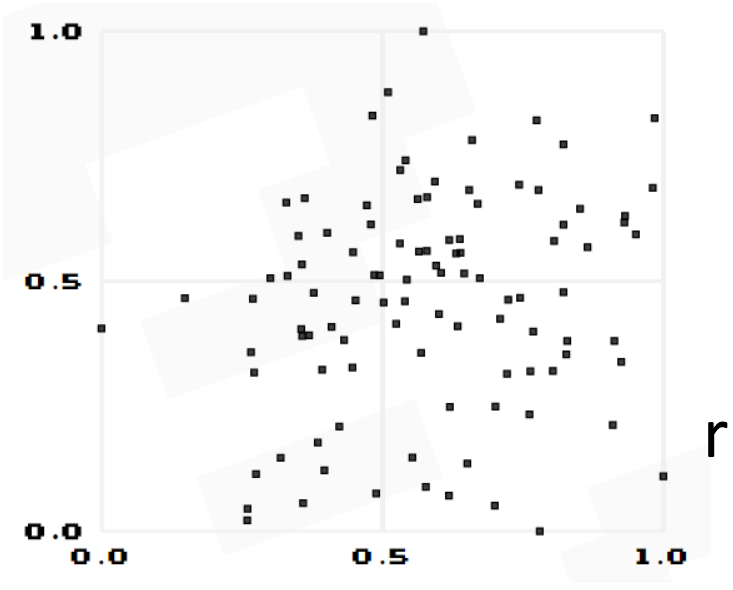
http://guessthecorrelation.com/

Before we begin: Guess the correlation!

r is the **correlation coefficient**. It is used to see how well a **simple** regression fits.

It goes from -1 to 1, and a value close to -1 or 1 means that the correlation between the two variables is high.





r = 0.2

Before we begin: Guess the correlation!

r is the **correlation coefficient**. It is used to see how well a **simple** regression fits.

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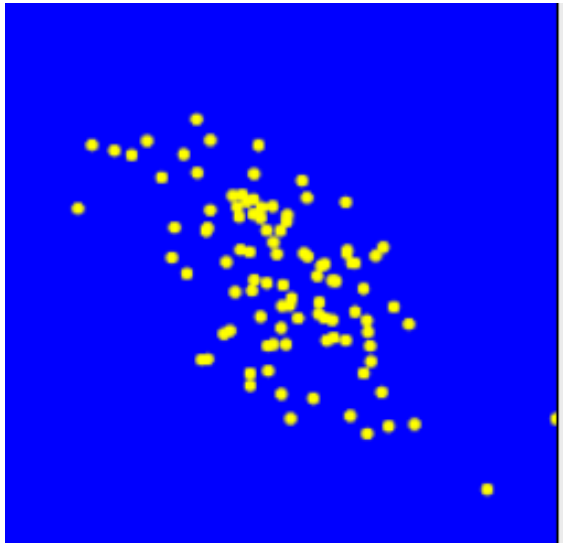
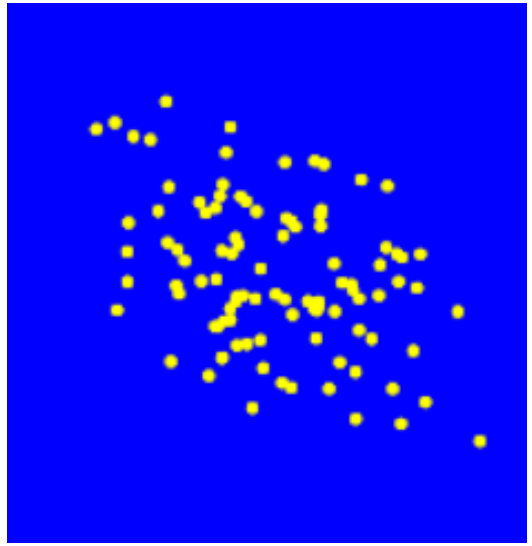
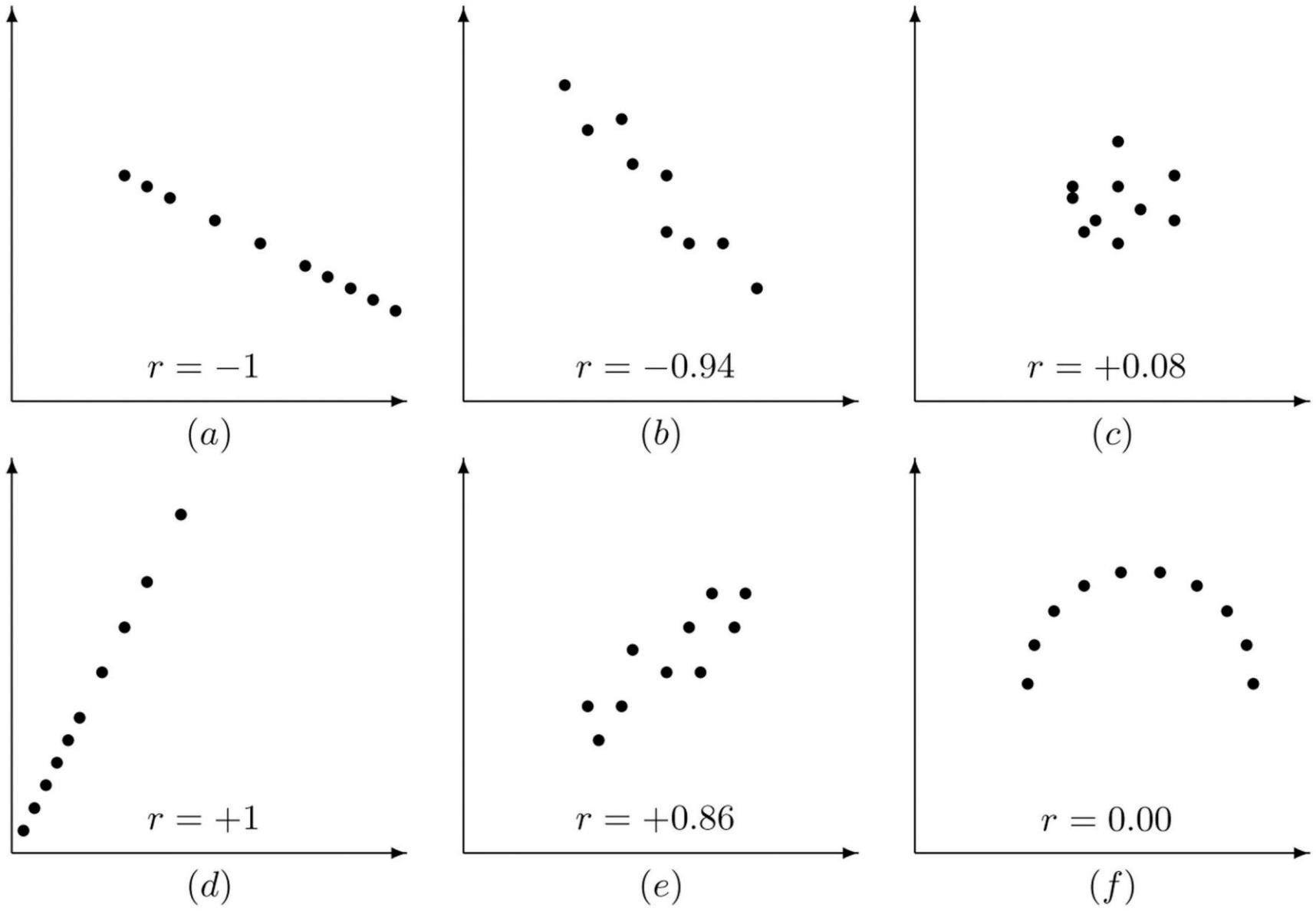


r = -0.61



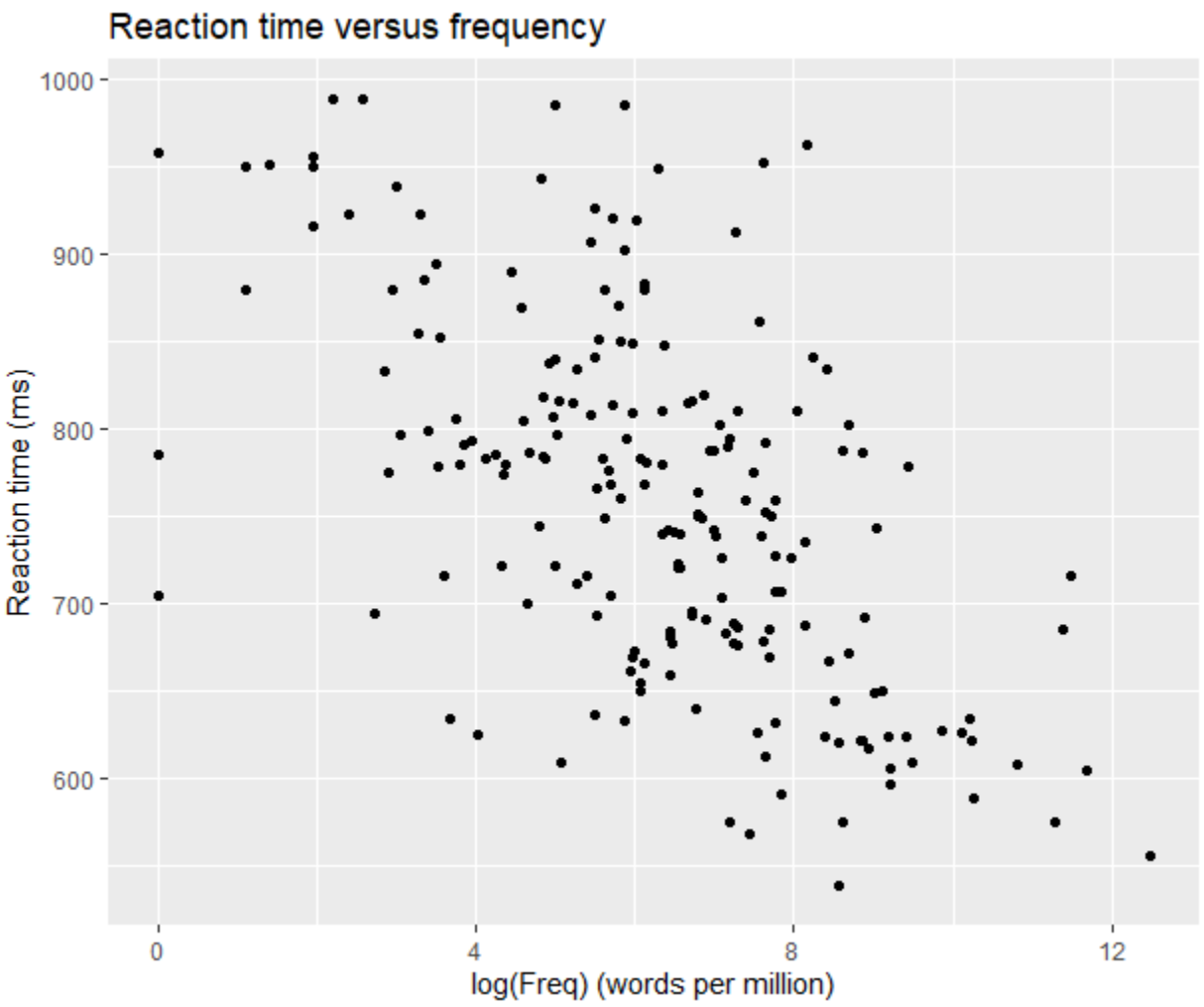
r = -0.38

Before we begin: Guess the correlation!



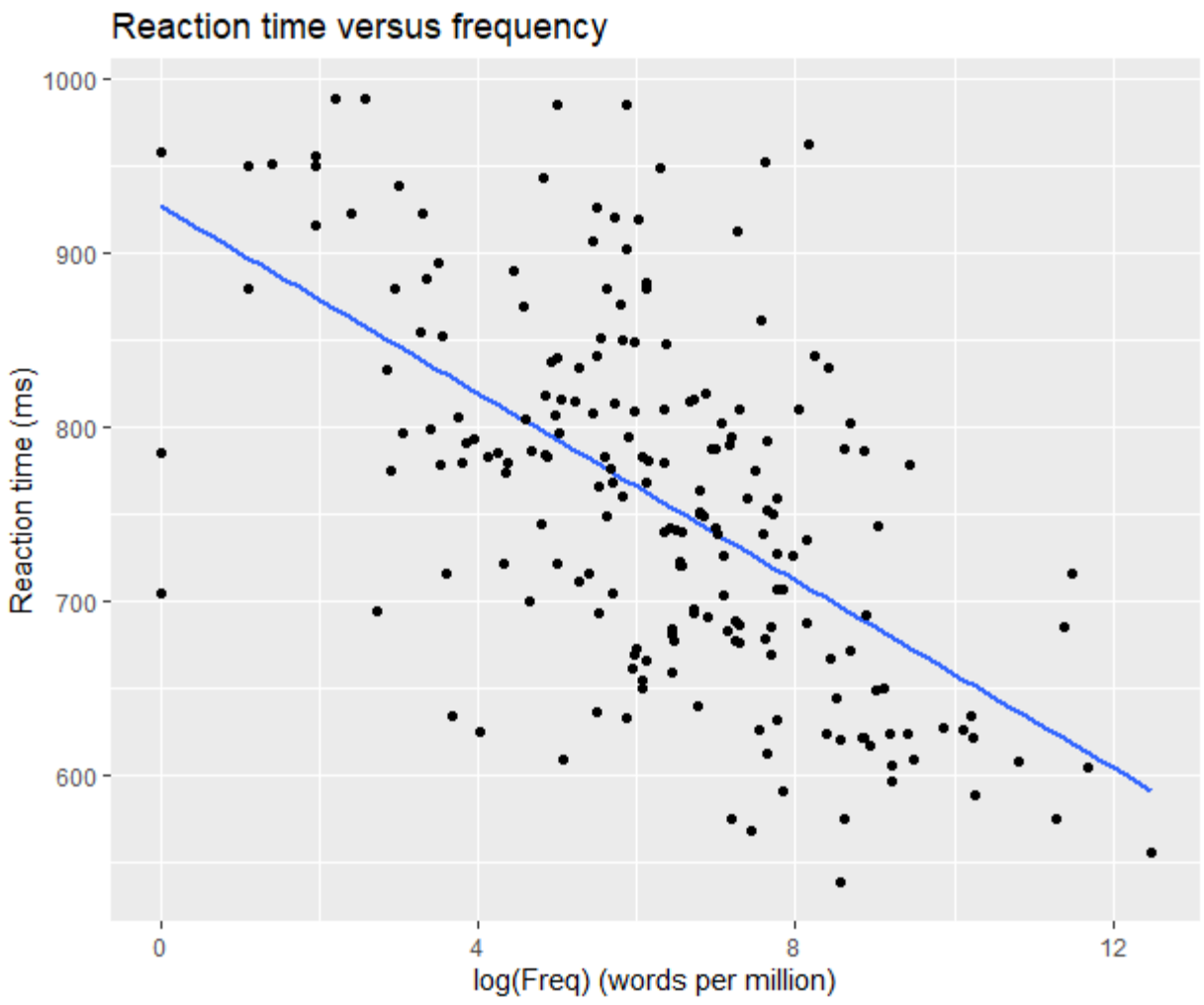
How to connect two numerical variables

If you have two quantitative variables, you can connect them with a **scatterplot**.
These dots are the **observations**. Scatterplots help you see patterns in the observations.



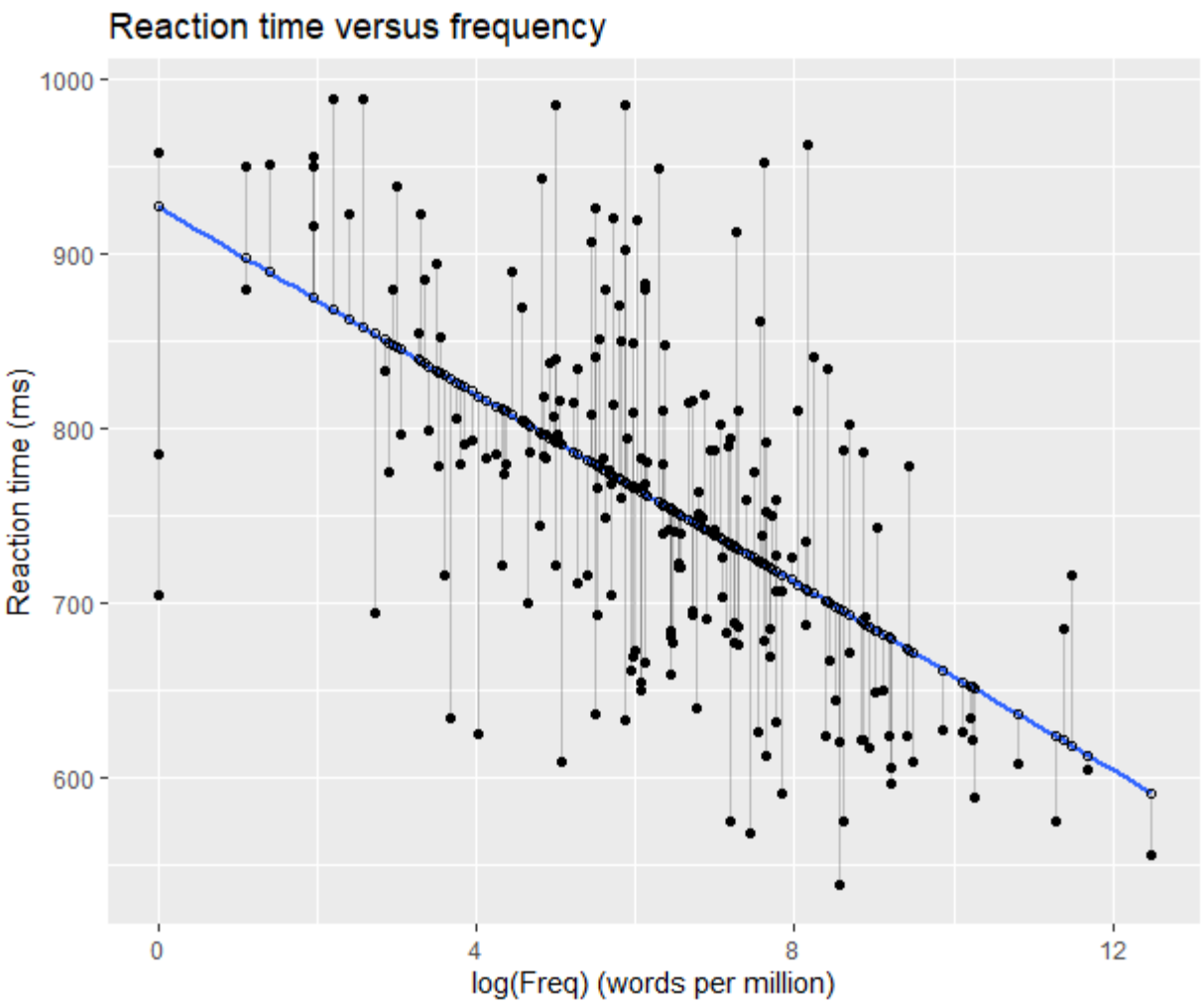
How to connect two numerical variables

Let's imagine a straight line that goes through as many of the dots as we can.
This is the **regression line**.



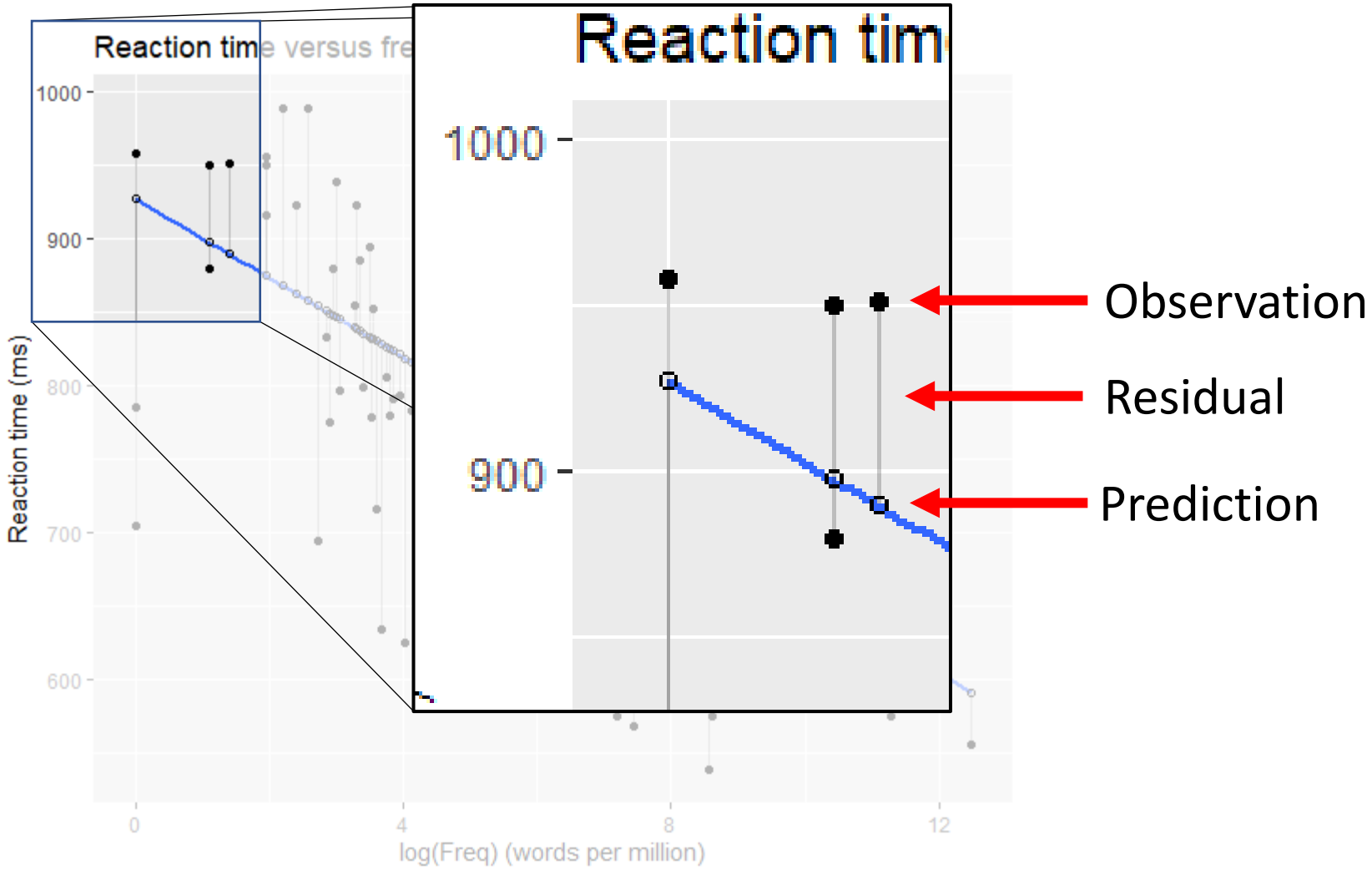
How to connect two numerical variables

Our preferred line is the one that **minimizes** the distance between all the dots and the line. These distances are the **residuals**.



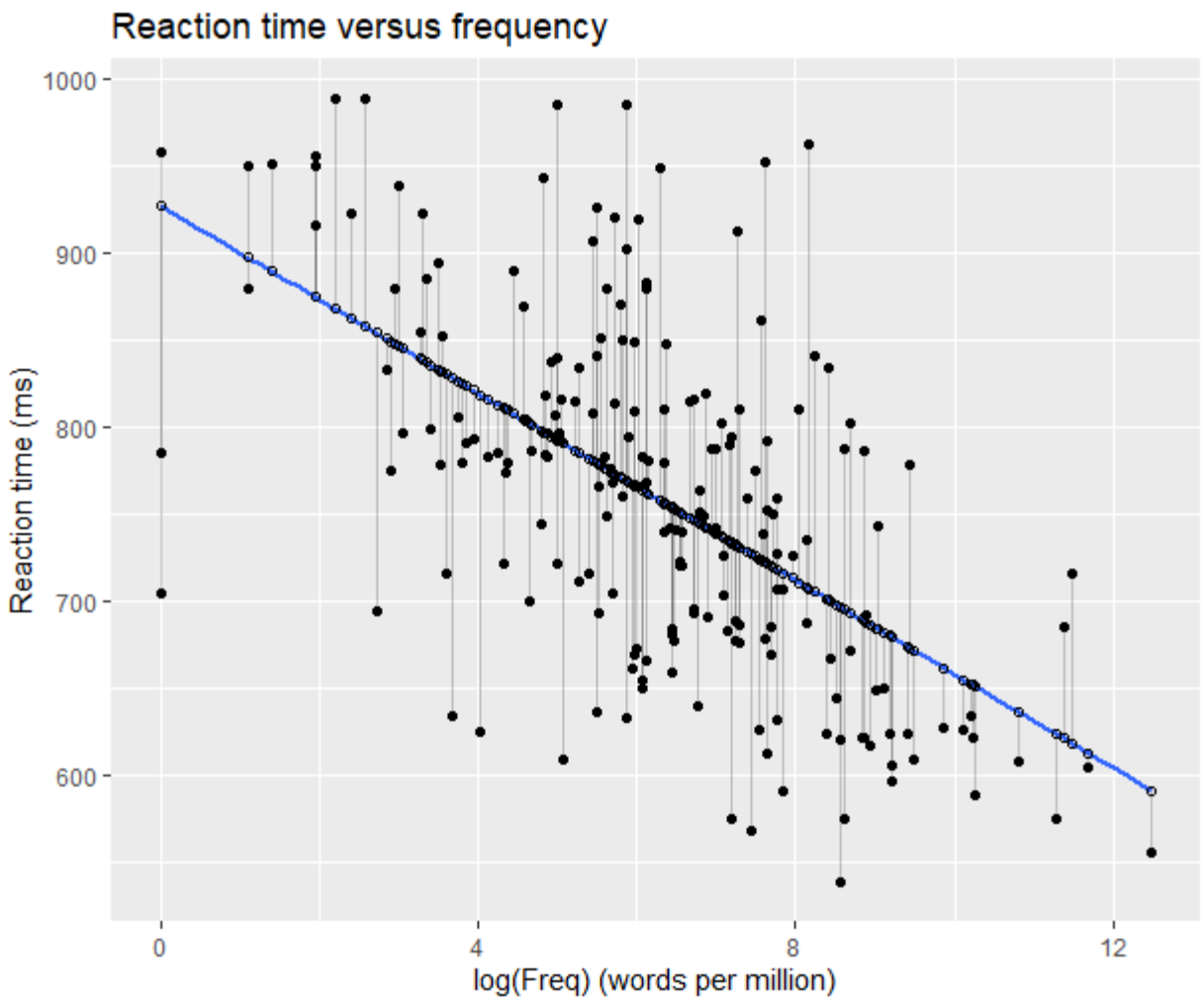
How to connect two numerical variables

The dots are the observations. The line has the **predictions**.



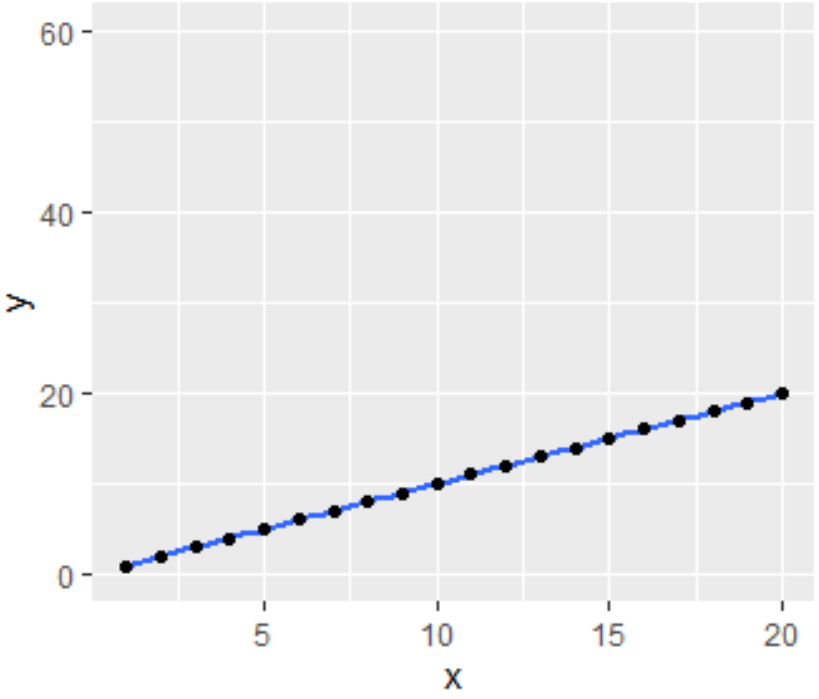
How to connect two numerical variables

Some residuals are positive, some are negative.
If you add them together, they should be approximately zero.

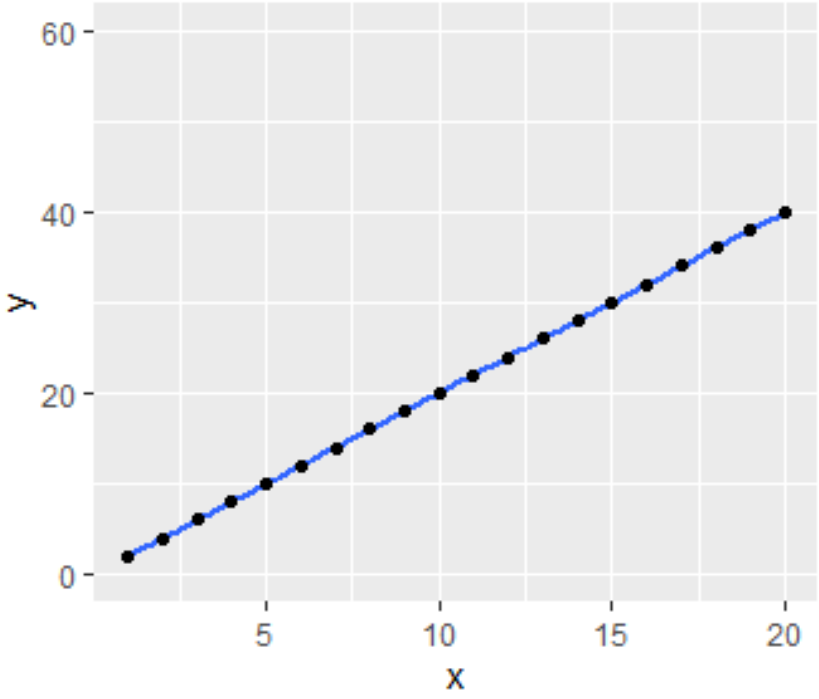


Basics of Regression

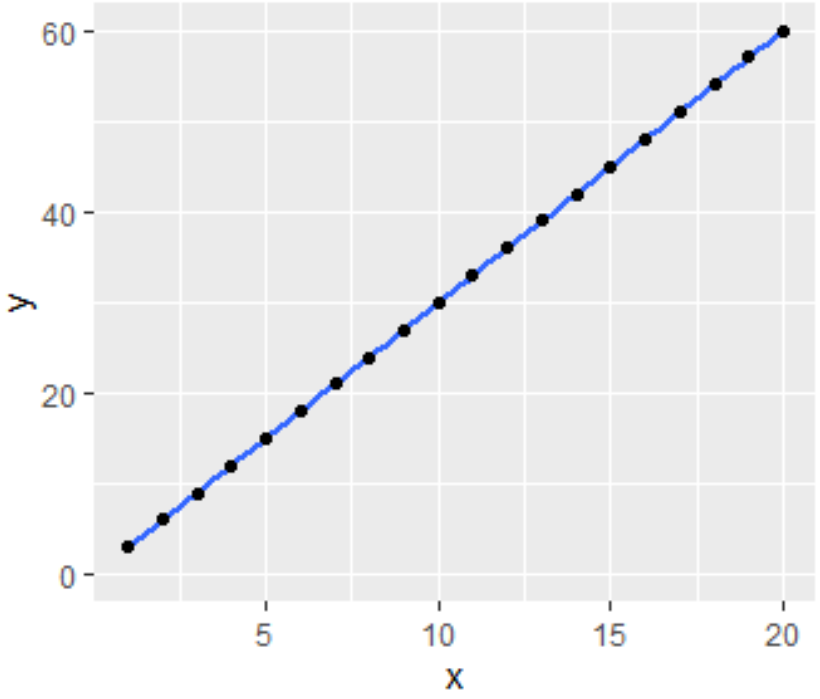
Your line can go up or down at different rates. We call that rate the **coefficient**.
It is represented with the letter **beta** (β).
If the coefficient is positive, it means that, as ‘x’ increases, ‘y’ also increases.



y = x
β = 1



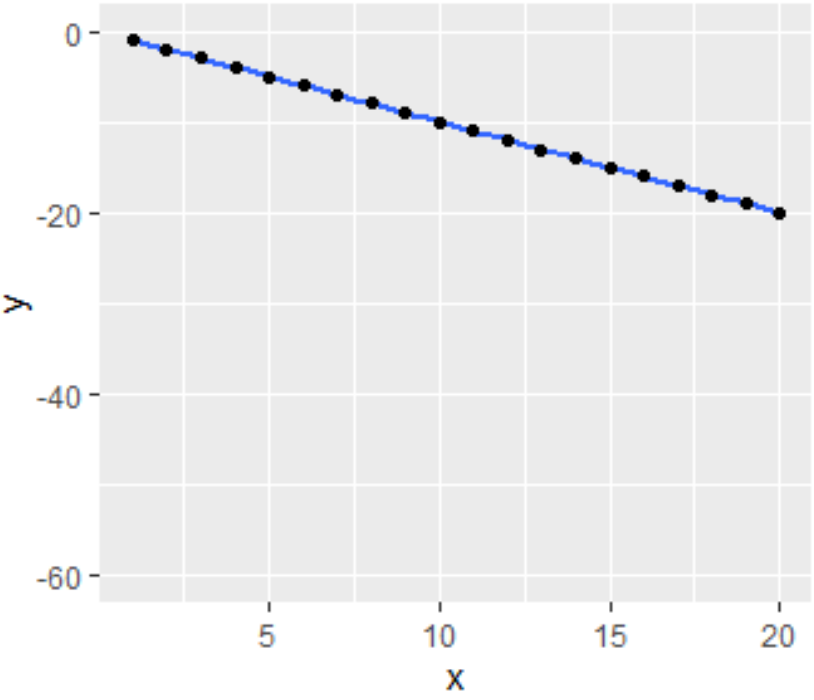
y = 2*x
β = 2



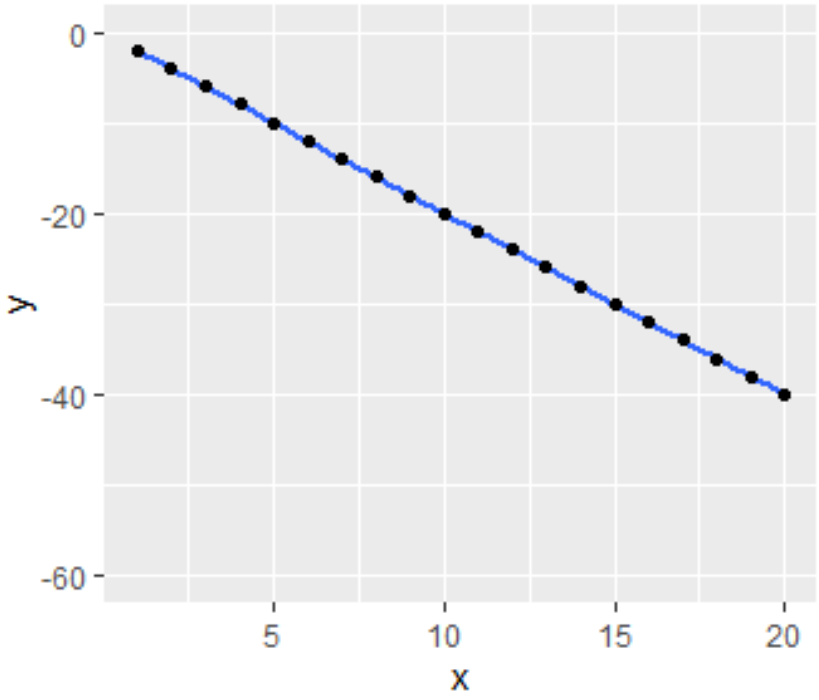
y = 3*x
β = 3

Basics of Regression

Coefficients can be positive or negative.
If they are negative, it means that, as ‘x’ increases, ‘y’ decreases.



y = -x
β = -1



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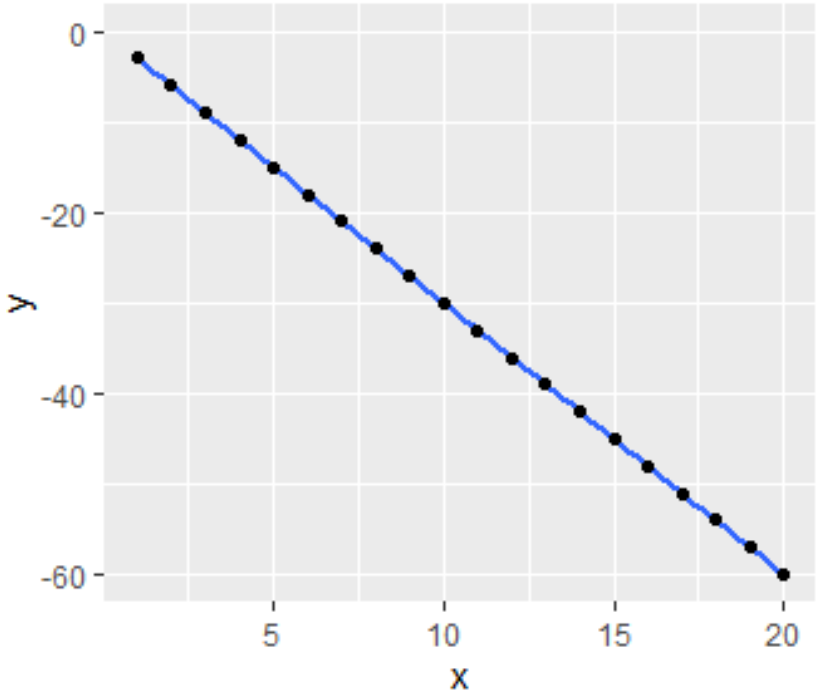
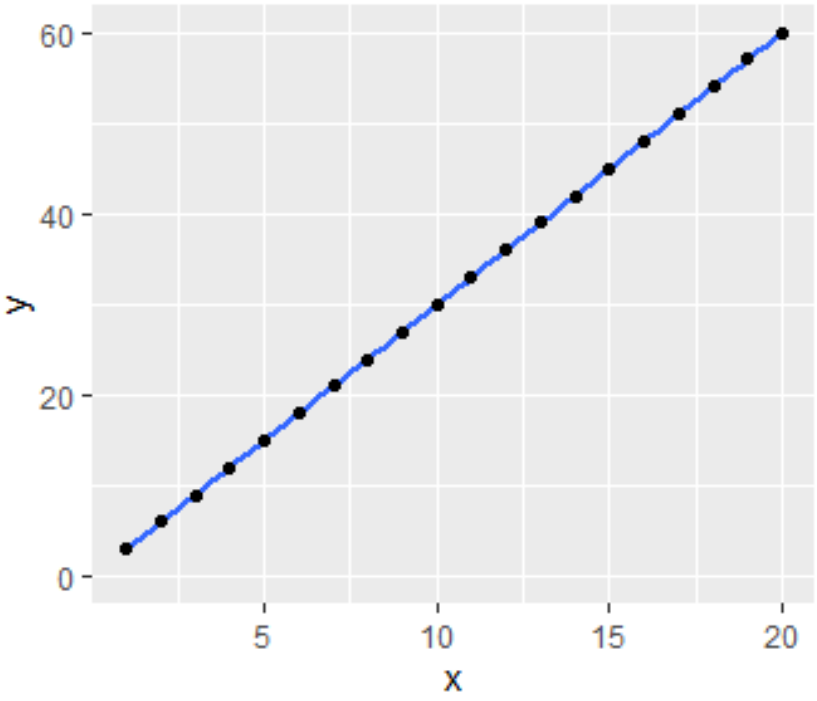


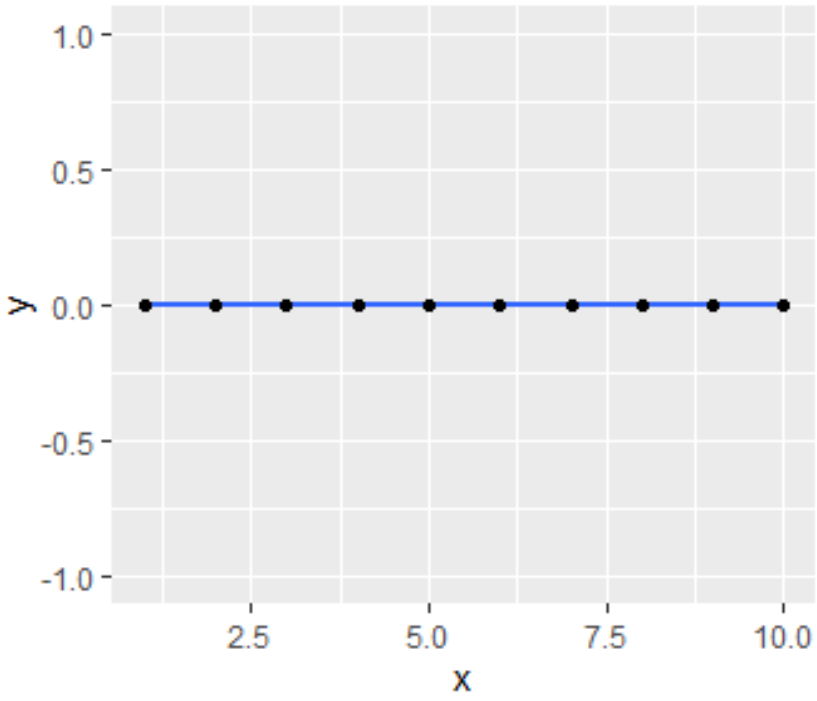
y = -3*x
β = -3

Basics of Regression

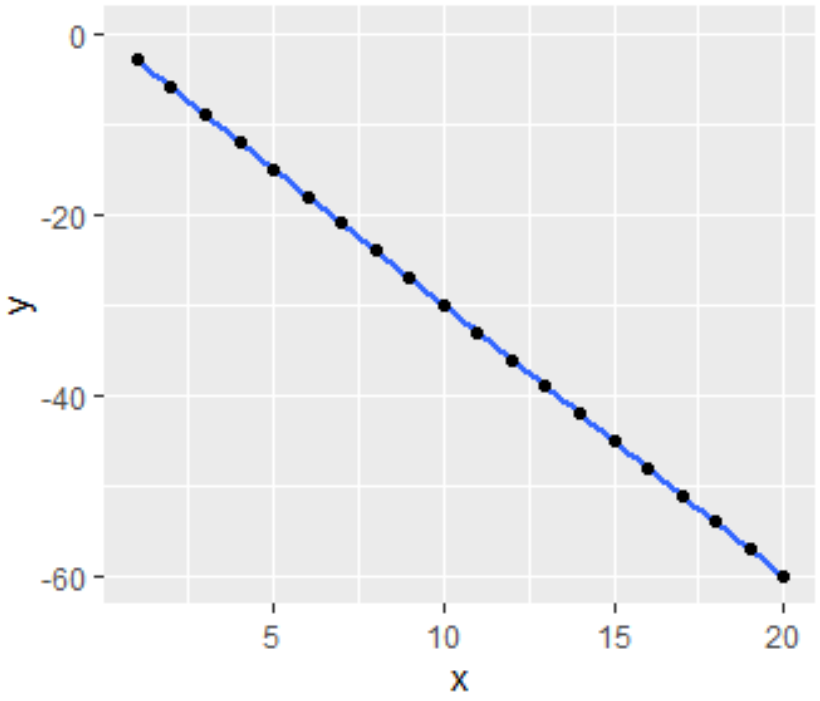
The coefficient can also be zero.
If the coefficient is zero, the value of ‘x’ has no influence on the value of ‘y’.



y = 3*x
β = 3



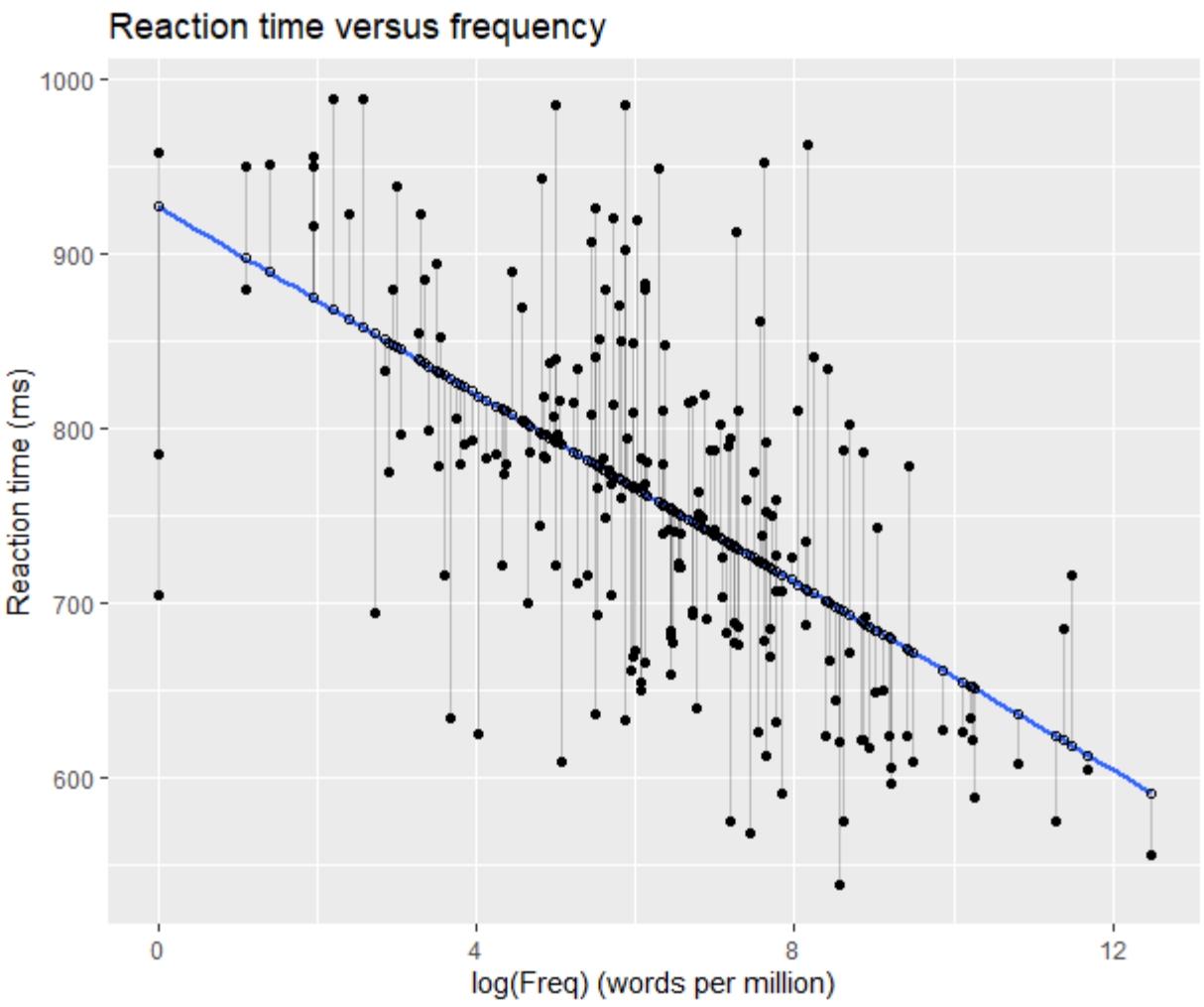
y = 0*x
β = 0



y = -3*x
β = -3

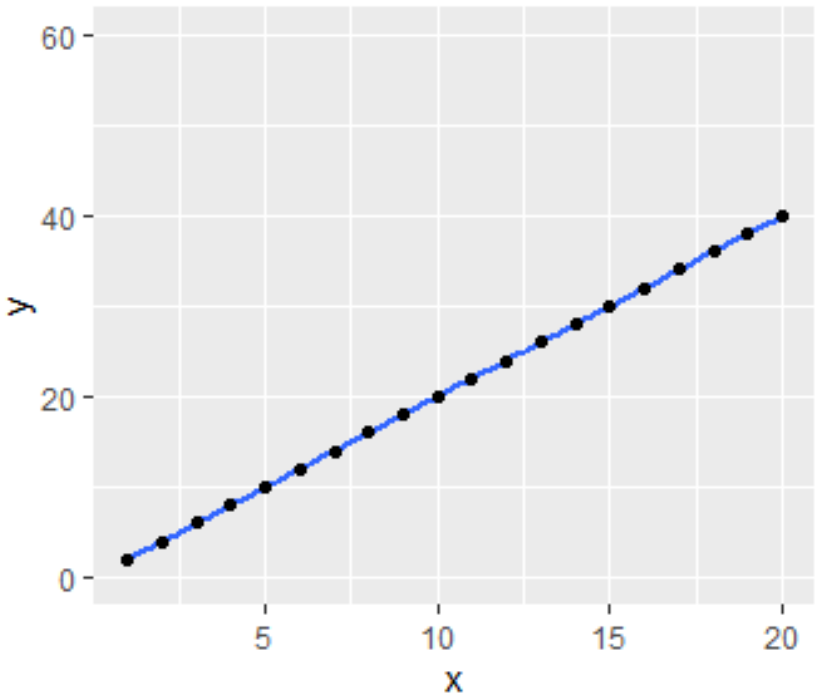
Basics of Regression

Is the coefficient positive or negative? Why?
Does this make linguistic sense?

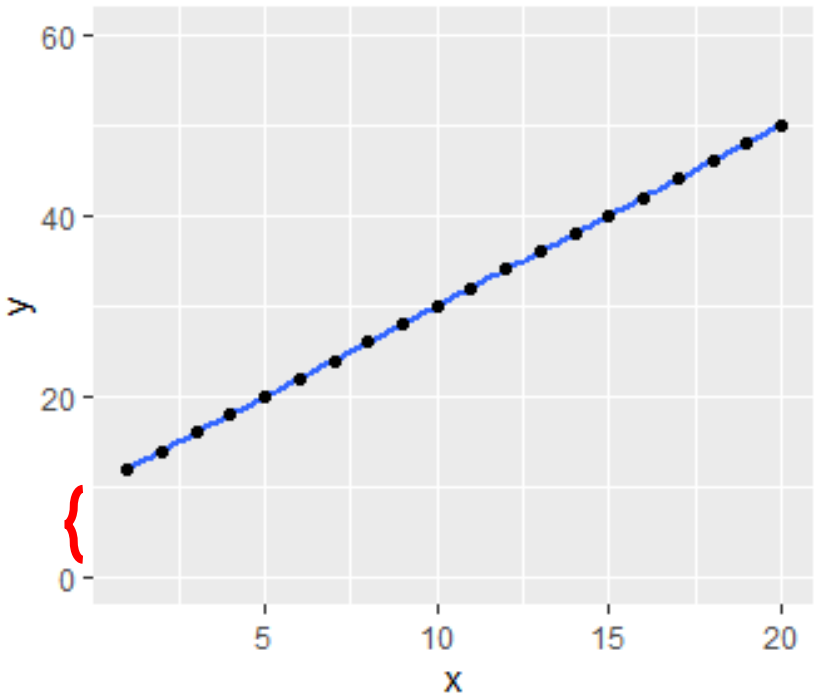


Basics of Regression

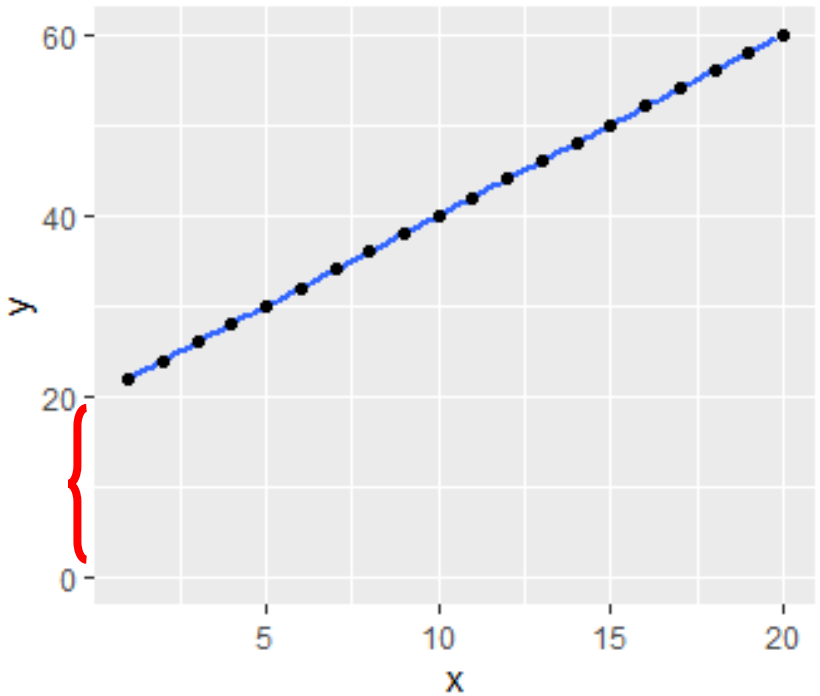
The **intercept** is the value that the equation takes when x=0.
(It "intercepts" the y-line).



y = 2*x + 0
β = 2
Intercept = 0



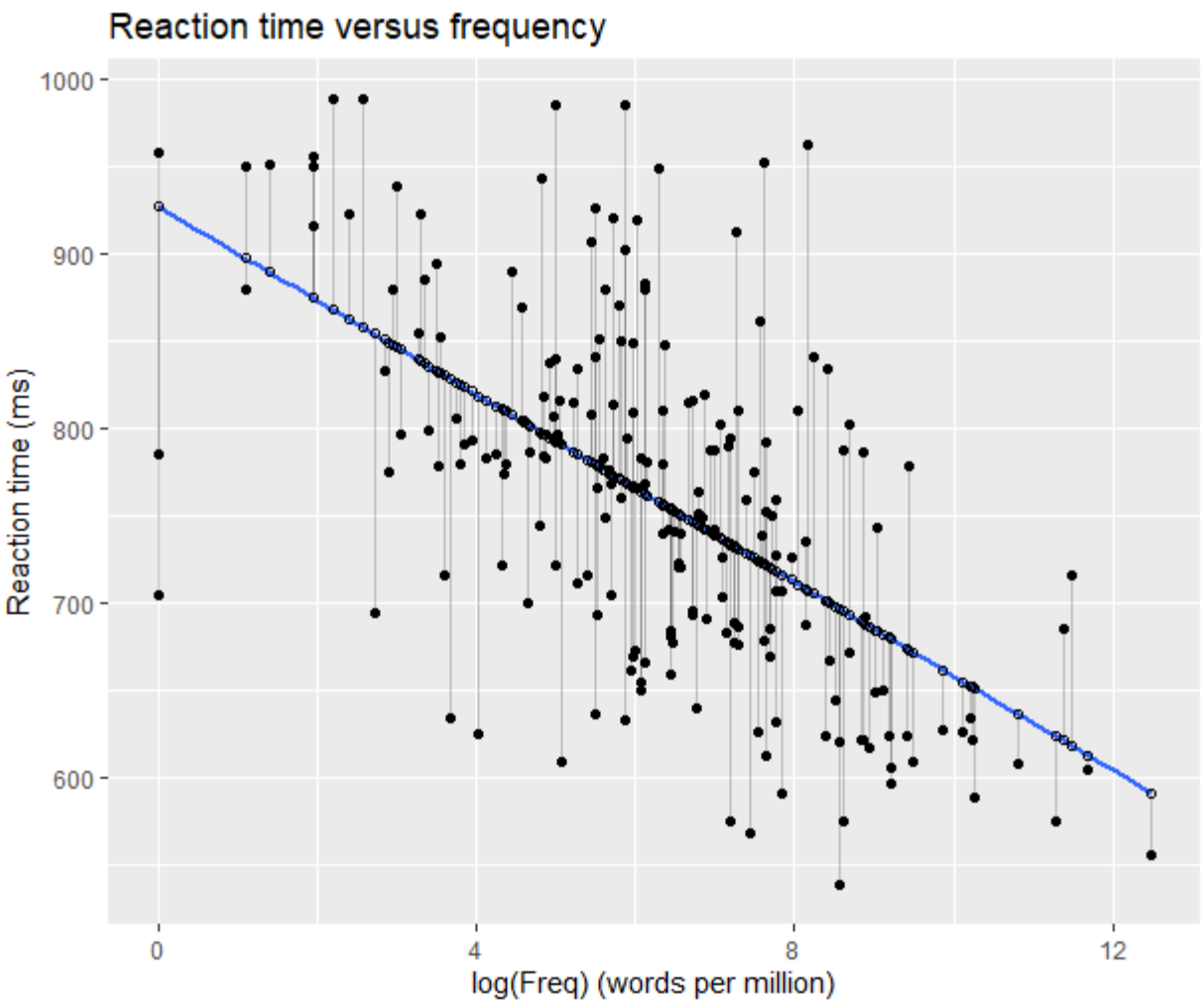
y = 2*x + 10
β = 2
Intercept = 10



y = 2*x + 20
β = 2
Intercept = 20

Basics of Regression

What is the intercept of this equation?
Does this make linguistic sense?



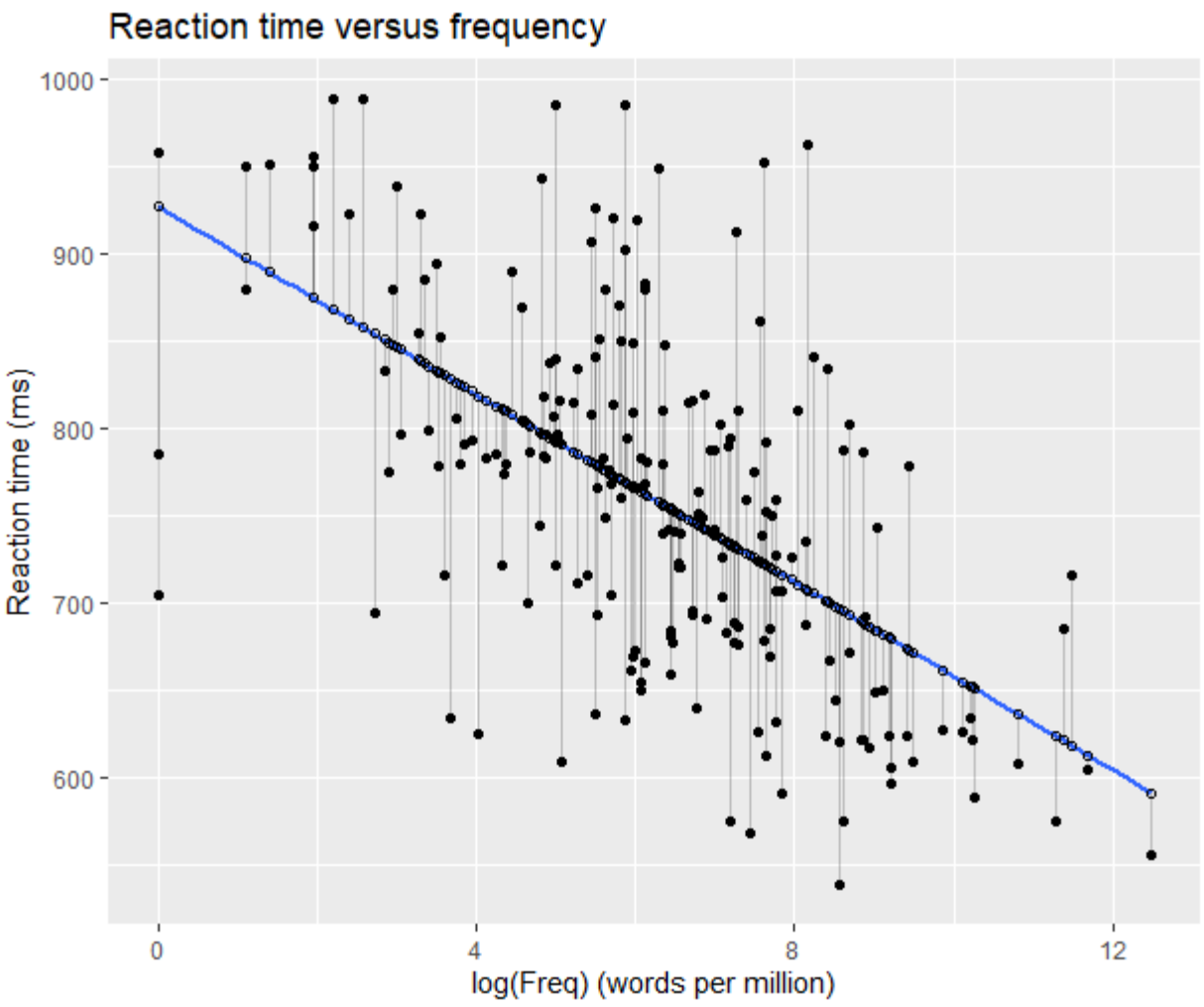
Basics of Regression

What is the intercept of this equation?
Does this make linguistic sense?

This intercept makes mathematical sense, but not linguistic sense.

Why? Because we don't have any real words where Freq=0.

This happens to intercepts and it's okay.



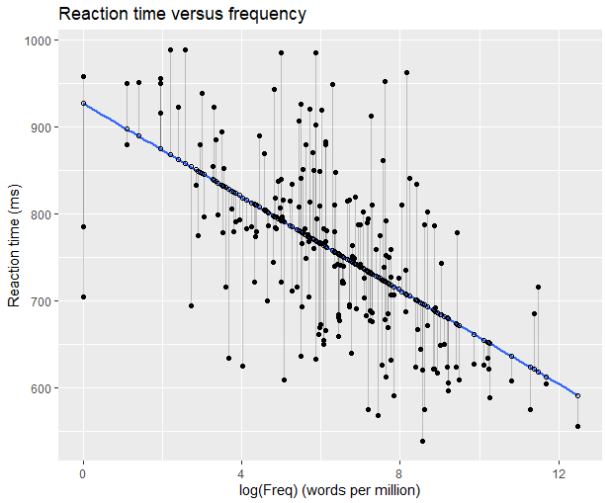
Summary so far

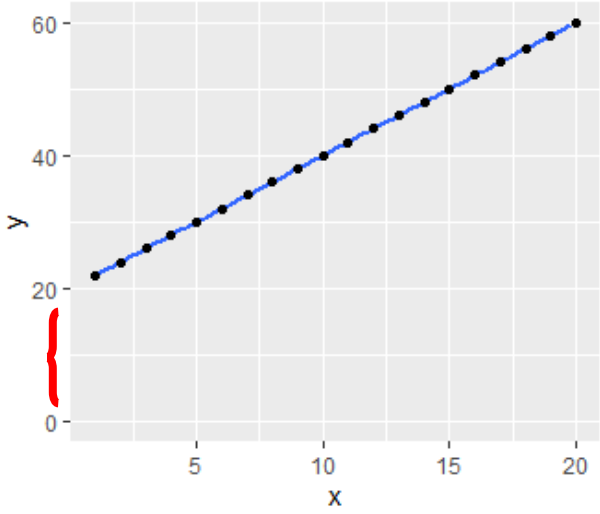
Regressions have:

A **coefficient** (the angle of the line). It can be positive, negative, or zero.

An **intercept** (where the line touches the ‘y’ axis) It is the value of ‘y’ when x=0.

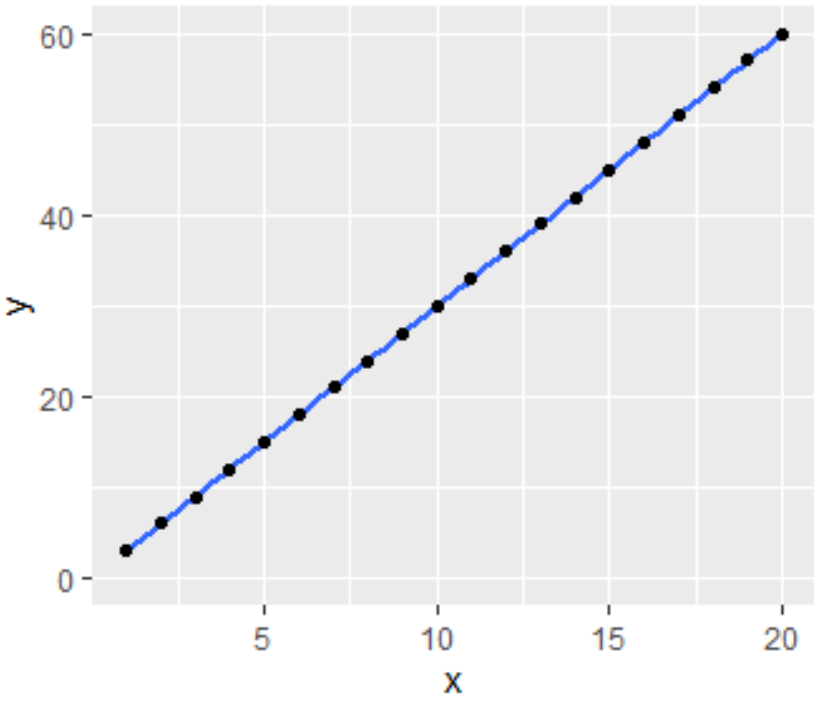
Residuals: The differences between the **predictions** (the line) and the **observations**.



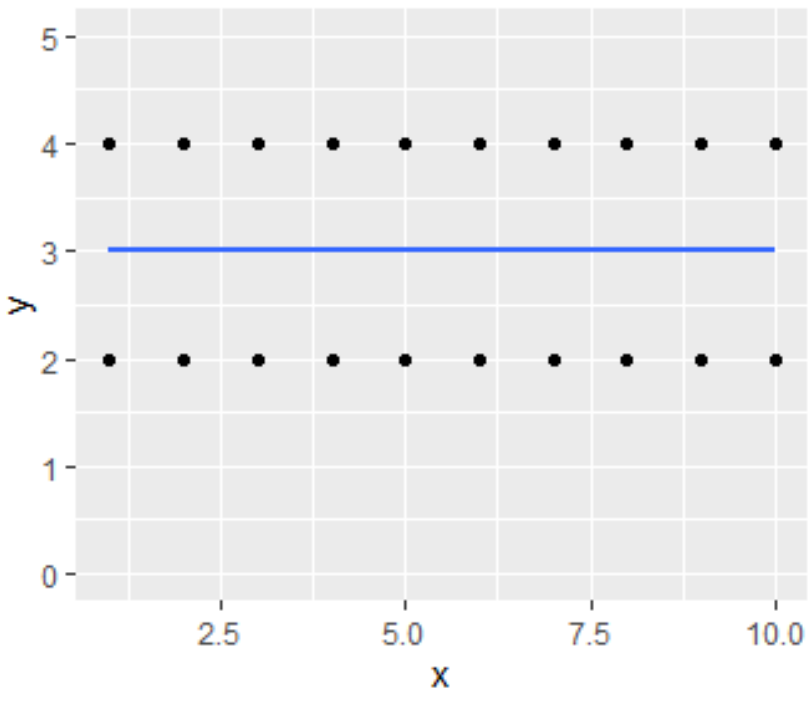


Basics of Regression

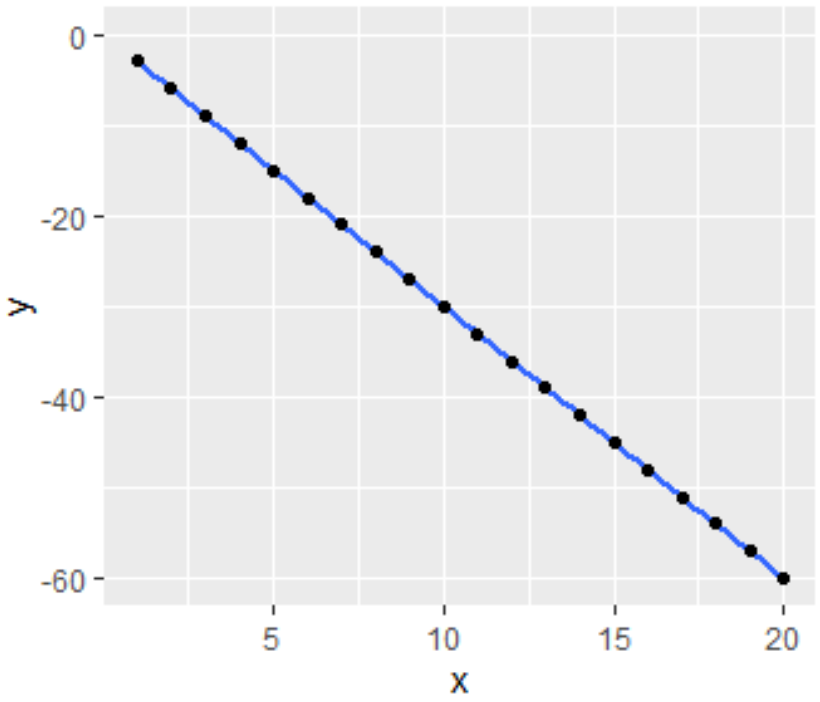
Let's go back to the correlation coefficient:



r = 1
All dots fit on the line perfectly.
The direction is positive.



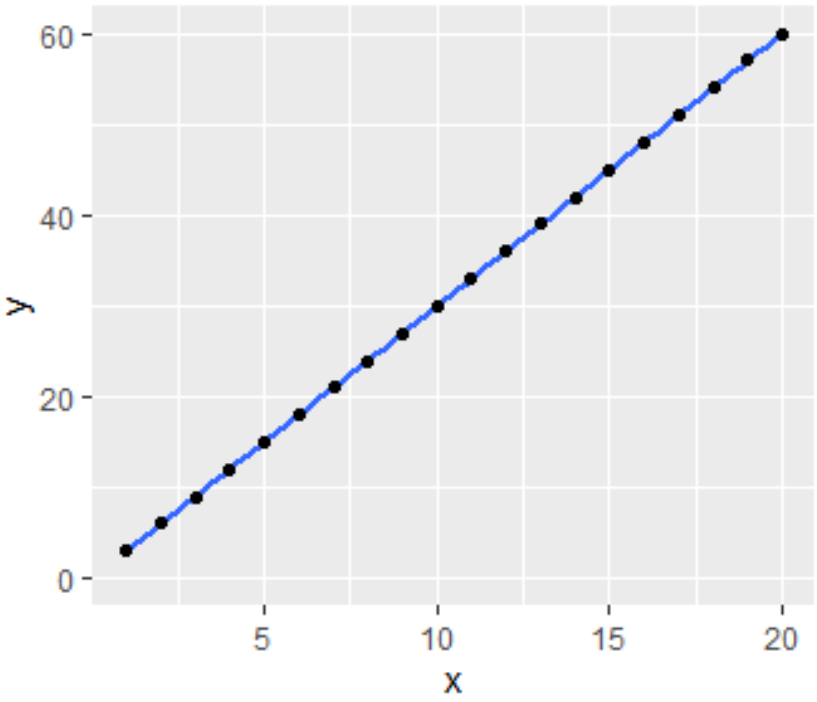
r = 0
None of the dots
fall on the line.



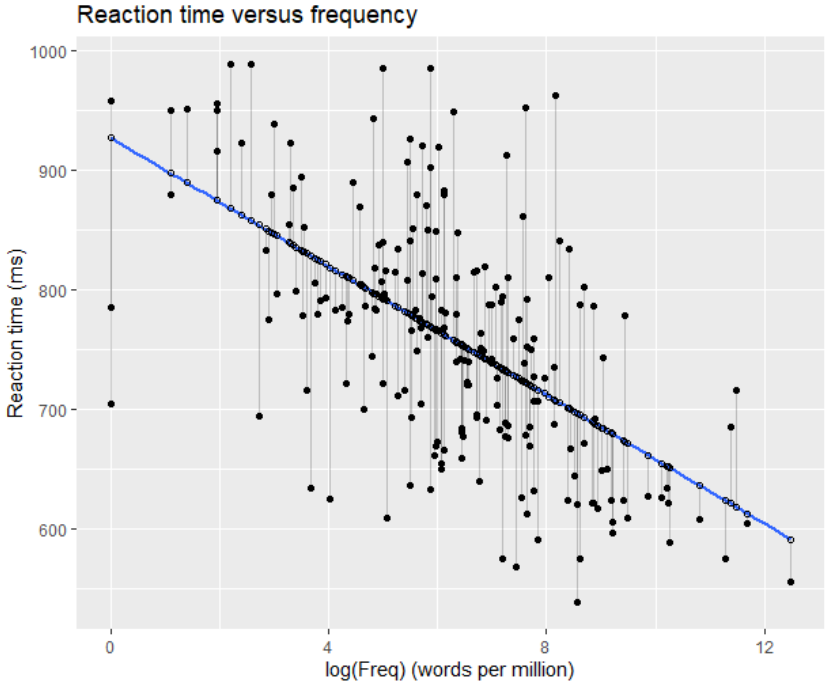
r = -1
All dots fit on the line perfectly.
The direction is negative.

Basics of Regression

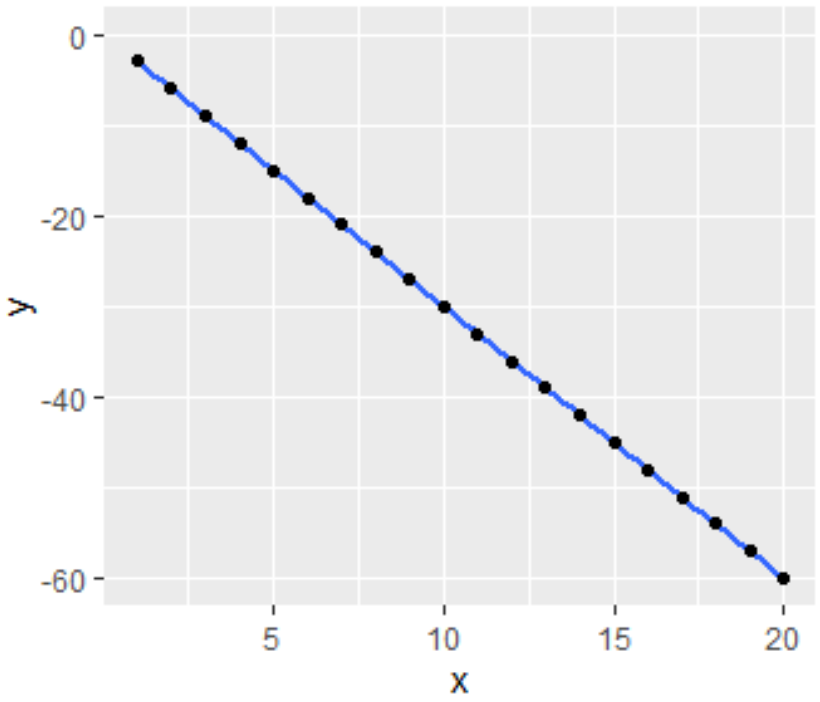
We use **R^2 (r-squared)** to indicate the goodness of fit, regardless of direction.
This is similar to our measures of effect size (eta-squared for ANOVA, phi for chi-squared)



$R^2 = 1$
All dots fit on the line perfectly.



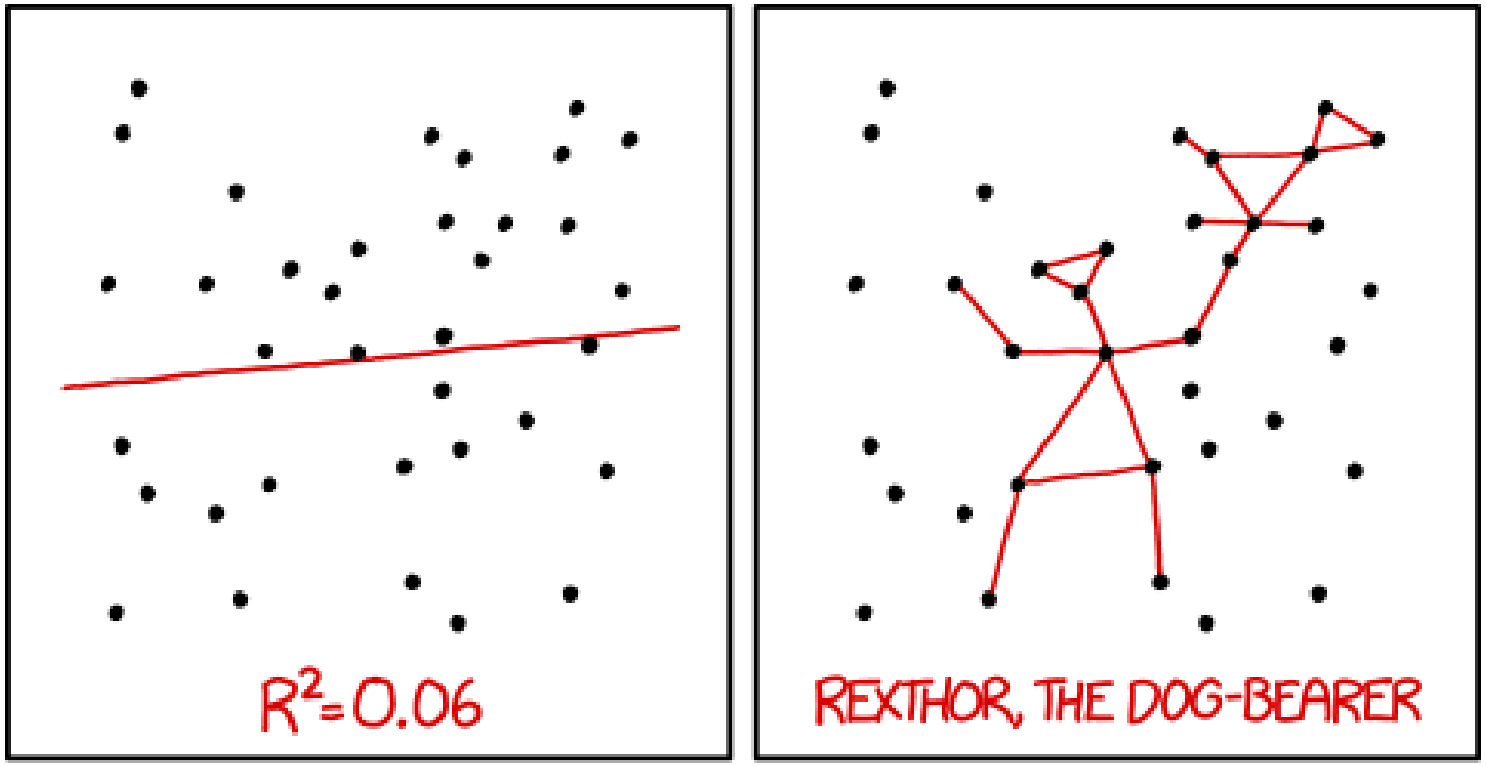
$R^2 = 0.34$
Some dots are on the line. There is a general negative trend.



$R^2 = 1$
All dots fit on the line perfectly. The direction is negative.

Basics of Regression

Despite the joke, R^2 can be small because humans are very multi-causal.
But always check that your R^2 is not too small.



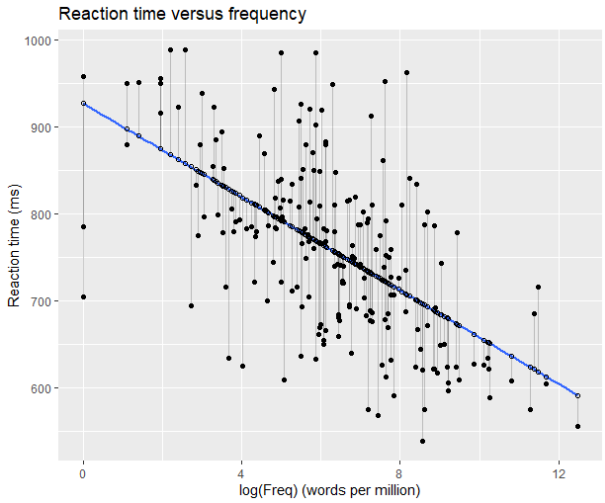
I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER TO GUESS THE DIRECTION OF THE CORRELATION FROM THE SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

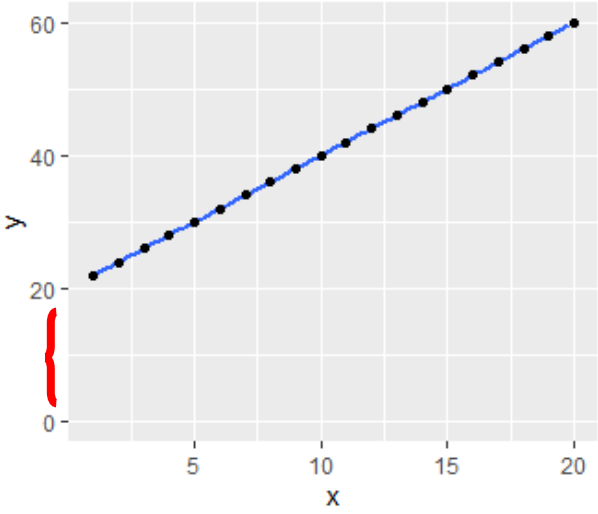
Summary so far

Regressions have:

Coefficient
Intercept
Residuals

R^2: Measurement of effect size

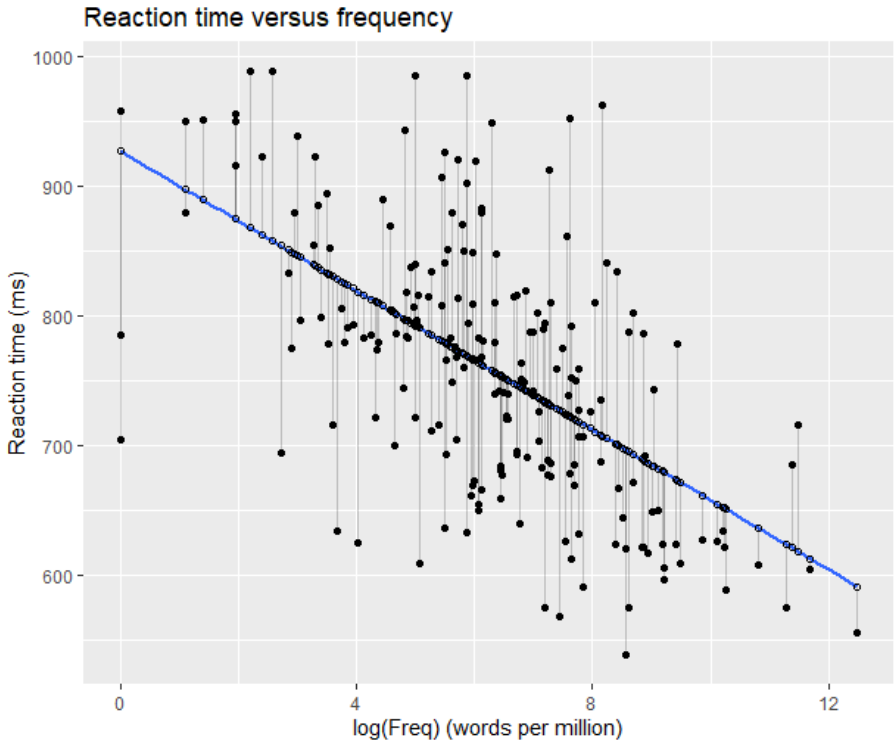




Regression and Linguistics

In English, the frequency of words is related to the reaction time when you see the words on the screen.

	Frequency numberOfWords	log(Frequency) log(numberOfWords)	Reaction time milliseconds
few	264606	12.49	555
rundown	700	6.55	720
fossilize	19	2.94	880



Regression and Linguistics

```
> elpFreqModel = lm(I_Mean_RT  ~ Log_Freq_HAL, data = elp)
> summary(elpFreqModel)

Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL, data = elp)

Residuals:
     Min       1Q   Median       3Q      Max 
-222.953  -53.734   -9.964   56.434  255.178 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)    926.77      17.68   52.42   <2e-16 ***
Log_Freq_HAL   -26.89       2.65  -10.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 84.56 on 198 degrees of freedom
Multiple R-squared:  0.3422,	Adjusted R-squared:  0.3389 
F-statistic:   103 on 1 and 198 DF,  p-value: < 2.2e-16
```

Regression and Linguistics

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```

The y-variable goes on the left.
The x-variables go on the right.

Regression and Linguistics

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```

The maximum residual is 255. This means that one of our predictions was off by 255ms.

The median of the residuals (the error of the predictions) is -10 ms.

50% of our predictions are off by about 55ms.

Regression and Linguistics

```
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Residual standard error: 84.56 on 198 degrees of freedom
Multiple R-squared:  0.3422,	Adjusted R-squared:  0.3389 
F-statistic:   103 on 1 and 198 DF,  p-value: < 2.2e-16
```

The intercept is 927 ms. It is significantly different from zero t(198)=52.4, p<0.0001.

The standard error is 18 ms. (This is the same standard error we used before: SE=SD/√n)

Internally, R is performing a t-test to see if the intercept is significantly different from zero.

Regression and Linguistics

```
> elpFreqModel = lm(I_Mean_RT  ~ Log_Freq_HAL, data = elp)
> summary(elpFreqModel)

Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL, data = elp)

Residuals:
     Min       1Q   Median       3Q      Max 
-222.953  -53.734   -9.964   56.434  255.178 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)    926.77      17.68   52.42   <2e-16 ***
Log_Freq_HAL   -26.89       2.65  -10.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 84.56 on 198 degrees of freedom
Multiple R-squared:  0.3422,	Adjusted R-squared:  0.3389 
F-statistic:   103 on 1 and 198 DF,  p-value: < 2.2e-16
```

The coefficient of logFreq is -27.

It is significantly different from zero: t(198)=-10.1, p<0.0001.

If the coefficient was zero, then logFreq would have no effect on the reaction time.

Regression and Linguistics

```
> elpFreqModel = lm(I_Mean_RT  ~ Log_Freq_HAL, data = elp)
> summary(elpFreqModel)

Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL, data = elp)

Residuals:
     Min       1Q   Median       3Q      Max 
-222.953  -53.734   -9.964   56.434  255.178 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)     926.77      17.68   52.42   <2e-16 ***
Log_Freq_HAL    -26.89       2.65  -10.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 84.56 on 198 degrees of freedom
Multiple R-squared:  0.3422,	Adjusted R-squared:  0.3389 
F-statistic:   103 on 1 and 198 DF,  p-value: < 2.2e-16
```

The equation is:
I_Mean_RT = 926.77 -26.9(Log_Freq_HAL)

For every increase of 1 unit in logFreq, there is a decrease of 27 units in RT.

Regression and Linguistics

What are the predicted values?

	log(Frequency) log(numberOfWords)	$Observed_{RT}$ ms	$Predicted_{RT}$ ms
few	12.49	555	
rundown	6.55	720	
fossilize	2.94	880	

```
I_Mean_RT = 926.77 - 26.9(Log_Freq_HAL)
```

Regression and Linguistics

What are the predicted values?

	log(Frequency) log(numberOfWords)	$Observed_{RT}$ ms	$Predicted_{RT}$ ms
few	12.49	555	591
rundown	6.55	720	751
fossilize	2.94	880	848

```
I_Mean_RT = 926.77 - 26.9(Log_Freq_HAL)
```

Regression and Linguistics

What are the predicted values?

	log(Freq)	Obs_{RT}	$Pred_{RT}$
few	12.49	555	591
rundown	6.55	720	751
fossilize	2.94	880	848

```
I_Mean_RT = 926.77 - 26.9(Log_Freq_HAL)
```

Regression and Linguistics

```
> elpFreqModel = lm(I_Mean_RT  ~ Log_Freq_HAL, data = elp)
> summary(elpFreqModel)

Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL, data = elp)

Residuals:
     Min       1Q   Median       3Q      Max 
-222.953  -53.734   -9.964   56.434  255.178 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)    926.77      17.68   52.42   <2e-16 ***
Log_Freq_HAL   -26.89       2.65  -10.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 84.56 on 198 degrees of freedom
Multiple R-squared:  0.3422,	Adjusted R-squared:  0.3389 
F-statistic:   103 on 1 and 198 DF,  p-value: < 2.2e-16
```

$R^2 = 0.34$

This is the effect size. The variable logFreq explains about 34% of the variance observed.

How to report a linear model

```
> elpFreqModel = lm(I_Mean_RT  ~ Log_Freq_HAL, data = elp)
Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL, data = elp)
Residuals:
     Min       1Q   Median       3Q      Max
-222.953  -53.734   -9.964   56.434  255.178
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    926.77      17.68   52.42   <2e-16 ***
Log_Freq_HAL   -26.89       2.65  -10.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 84.56 on 198 degrees of freedom
Multiple R-squared:  0.3422, Adjusted R-squared:  0.3389
F-statistic:   103 on 1 and 198 DF,  p-value: < 2.2e-16
```

A linear regression was run on 200 English words, with the log-transformed frequency as the independent variable and the reaction time in milliseconds as the dependent variable. The reaction time decreases as the frequency increases. This effect is significant (β=-27, SE=2.6, t(198)=-10.1, p<0.0001, R^2=0.34).

Practice

In theory, familiarity is a proxy for frequency. These two values should be correlated.

Are familiarity and frequency correlated for the animal words in the `animals` dataset?

Assumptions

<u>Linear regression models have several assumptions</u>

The data should be randomly selected.

The y variable is numeric.

Both variables are roughly normal, and their relationship is roughly linear.

The residuals are homoscedastic (roughly the same for all values of ‘x’)

The residuals are normally distributed (we see this in the QQ-Plot)

The residuals are independent (they are not “autocorrelated”)

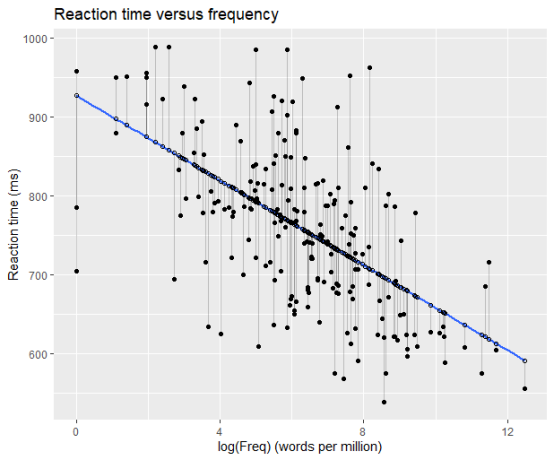
Assumptions

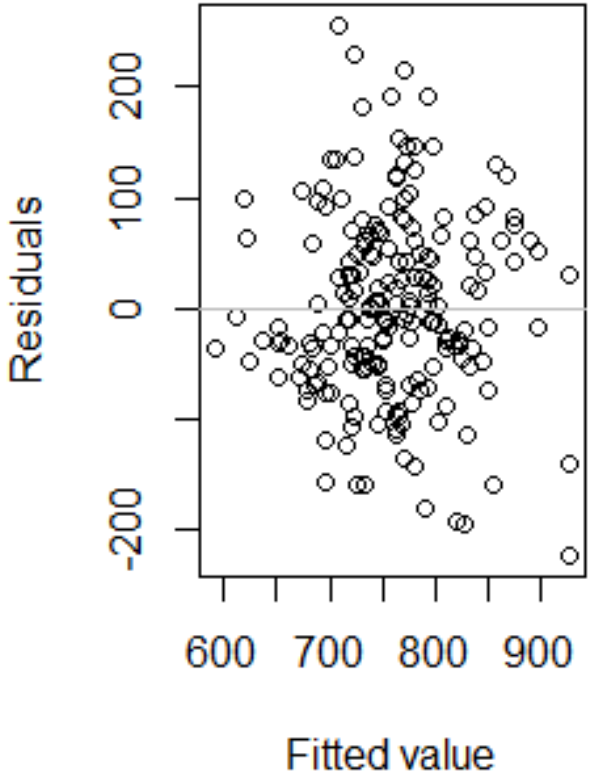
The residuals should be normally distributed:

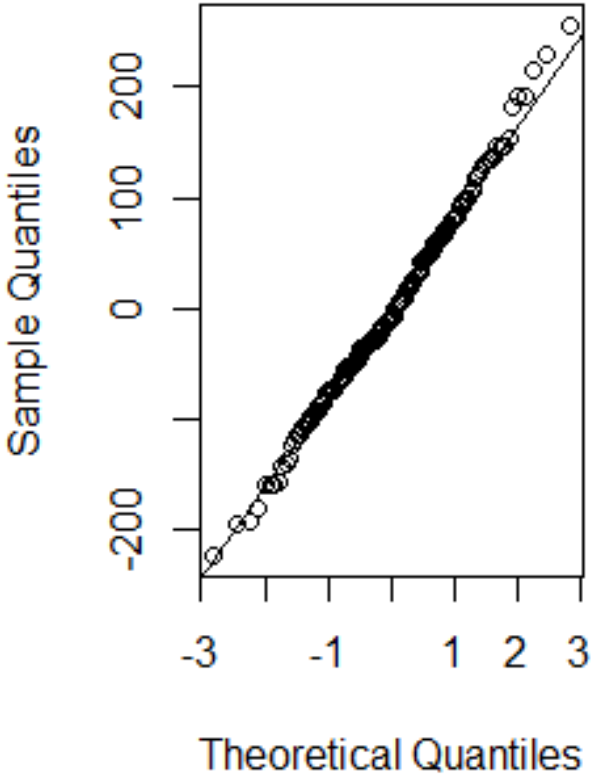
(1) When you plot them against the fitted values, they look like a cloud.

(2) When you make the QQ-Plot, the dots fall on the line.

Data for the ELP English Frequency Example

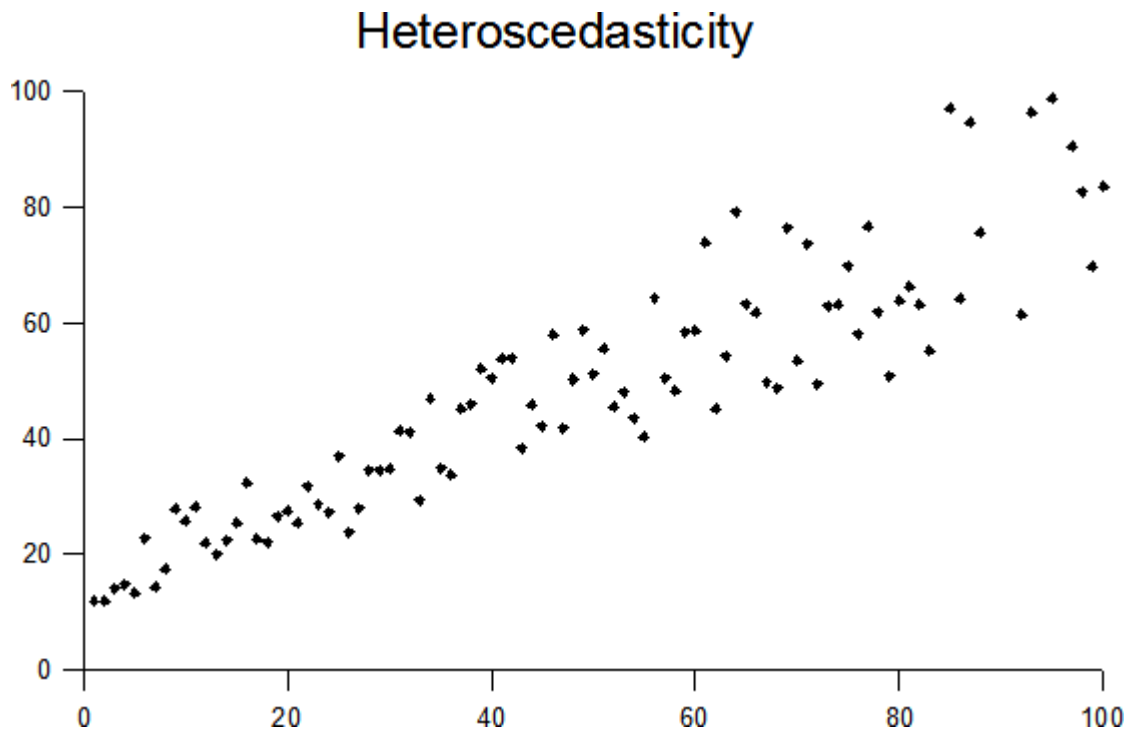


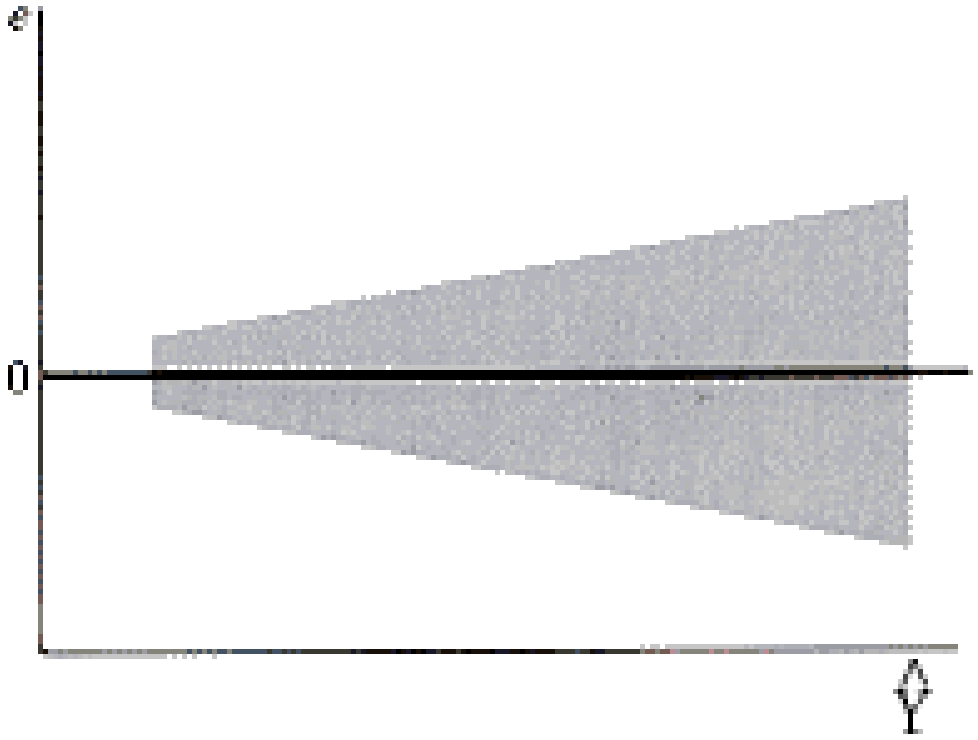




Assumptions

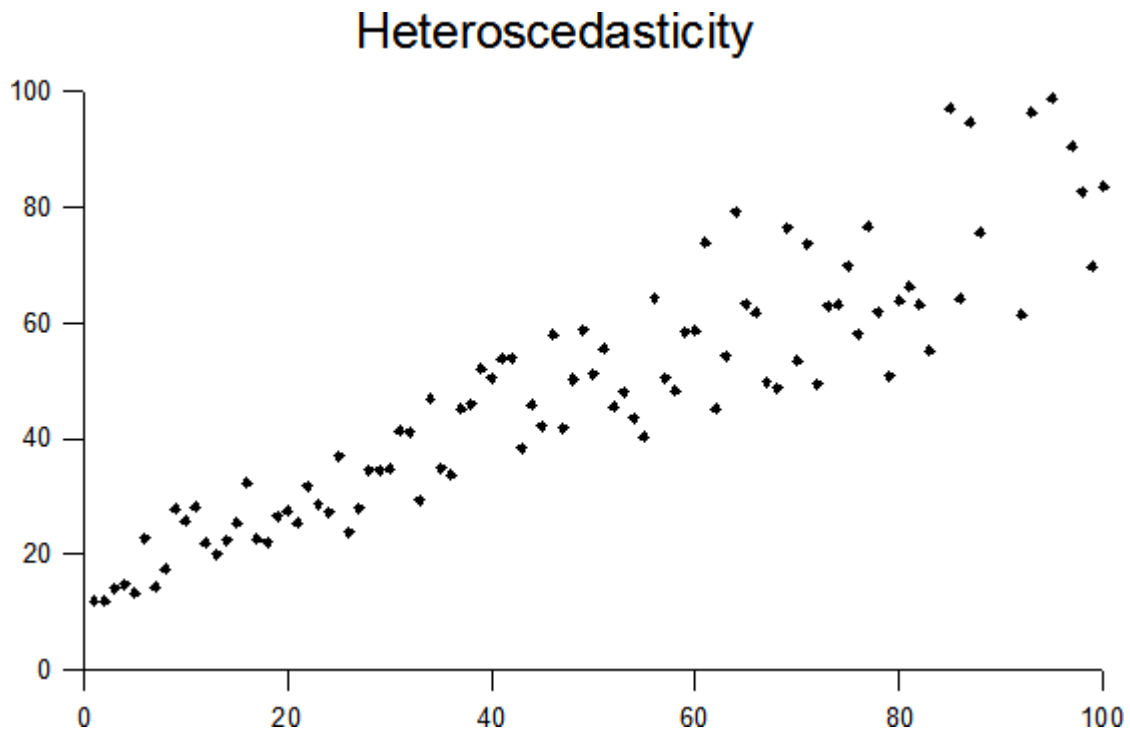
Funnels in the residuals indicate that the values of the residuals are not the same for all of the ‘y’ values. This violates our assumptions.

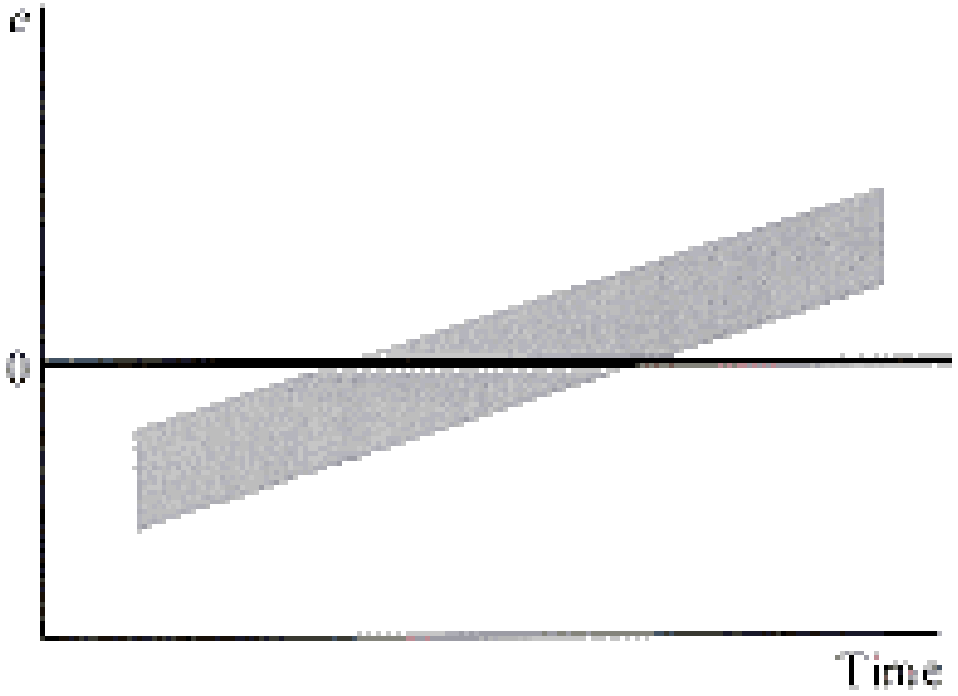




Assumptions

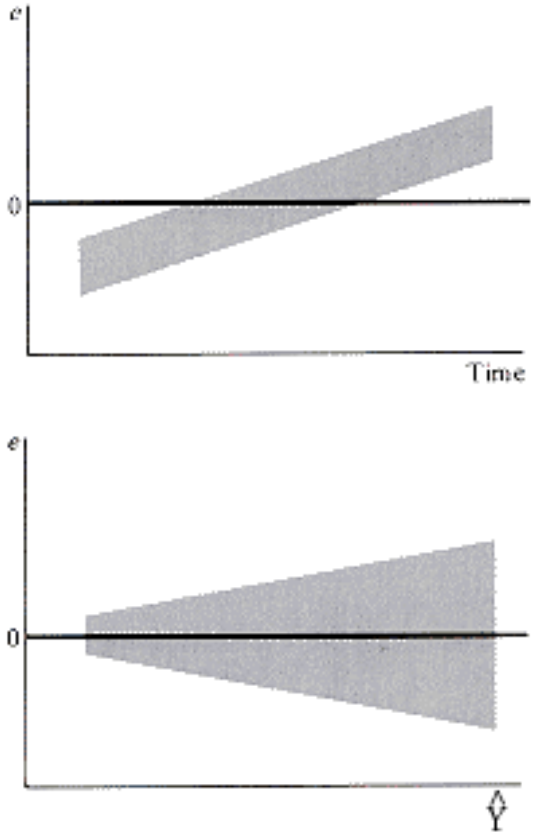
Residuals that have a clear direction also violate our assumptions.
Good residuals should be roughly random.

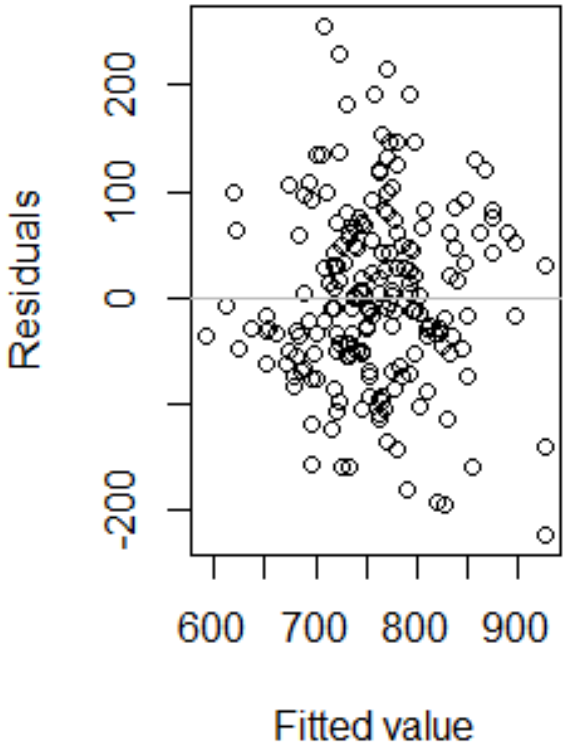




Assumptions

The patterns on the left are bad. The pattern on the right is good! The residuals look like a cloud. There is no discernible shape or direction in them.





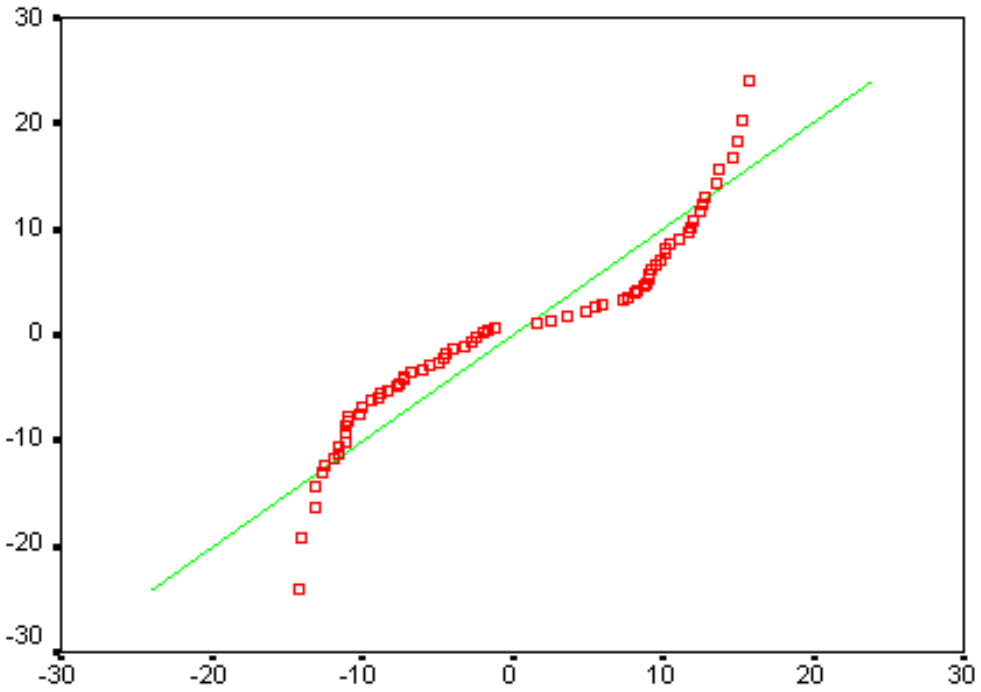
Heteroscedastic 👎

Homoscedastic 👍

Assumptions

The residuals should be independent. They should not be **autocorrelated**.

(Things that are autocorrelated depend on the previous value. For example, temperatures are autocorrelated. They have cycles, and today's temperature depends on yesterday's temperature).



Assumptions

We have formal tests for these:

```
> shapiro.test(elpFreqModel$residuals) # Normality of residuals
     Shapiro-Wilk normality test
data:  elpFreqModel$residuals
W = 0.99396, p-value = 0.5943

> ncvTest(elpFreqModel) # Same variance throughout
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 2.049159, Df = 1, p = 0.15229

> durbinWatsonTest(elpFreqModel) #AutoCorrelation
 lag Autocorrelation D-W Statistic p-value
   1     -0.09585457      2.186411    0.19
 Alternative hypothesis: rho != 0
```

Practice: Assumptions

Now you try it on the example! Does the `animal` example meet the assumptions?

Assumptions

Beware!

These are **ASSUMPTIONS**. They are not iron-clad rules.
In many cases we will asume that are data is normal for several reasons:

(1) Tradition. The experimental paradigm we are using regularly assumes its phenomenon is normal.

(2) Ease of use. Techniques like LM are fairly easy and are resilient to deviations of normality. That's why they are taught to scientists who are not statisticians. (Dartmouth does have a statistical consultant!)

You should always check whether it would be **reasonable** to assume that your distribution is normal.

Assumptions

You should always check whether it would be **reasonable** to assume that your **residuals** are normal.

Residuals that are reasonably normal:

- Look nebulous in the fitted values plot
- Fall roughly along the regression line in the Q-Q plot
- Most of the points are “average” and there aren’t many outliers

If you are worried, ask a statistician.
Or even better, go meet with them as you are designing your experiment.

Assumptions

You should always check whether it would be **reasonable** to assume that the **distribution of your variable** is normal.

A **variable** that is reasonably normal:

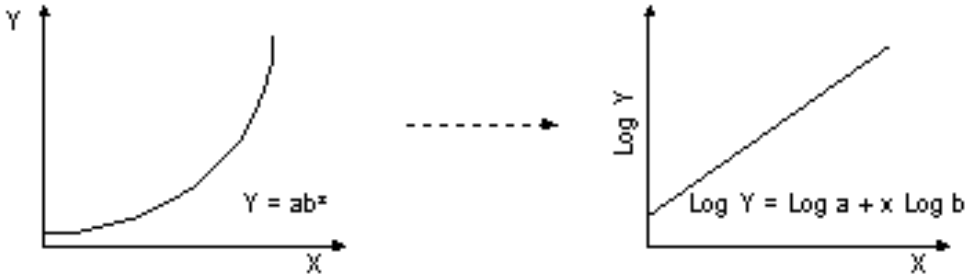
- Has one predominant mode
- Is fairly symmetrical
- Most of the points are “average” and there aren’t many outliers

If you are worried, ask a statistician.
Or even better, go meet with them as you are designing your experiment.

Assumptions

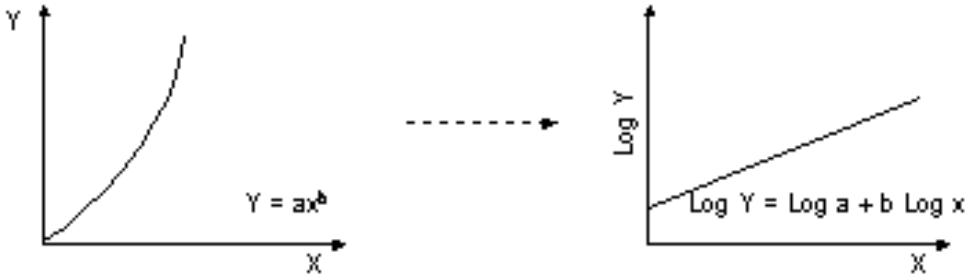
The relationship between the two variables should be roughly linear.
It's okay to apply transformations, but sometimes it will not work.

Polynomial:
$y = ab^x$



$\log(y) = \log(a) + x\log(b)$

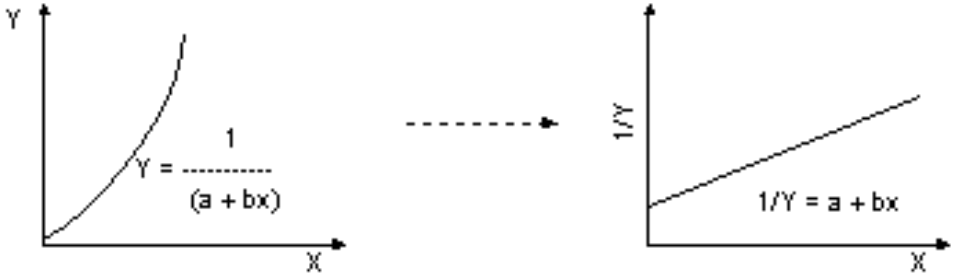
Exponential
$y = ax^b$



$\log(y) = \log(a) + b\log(x)$

Inverse:

$$y = \frac{1}{(a + bx)}$$



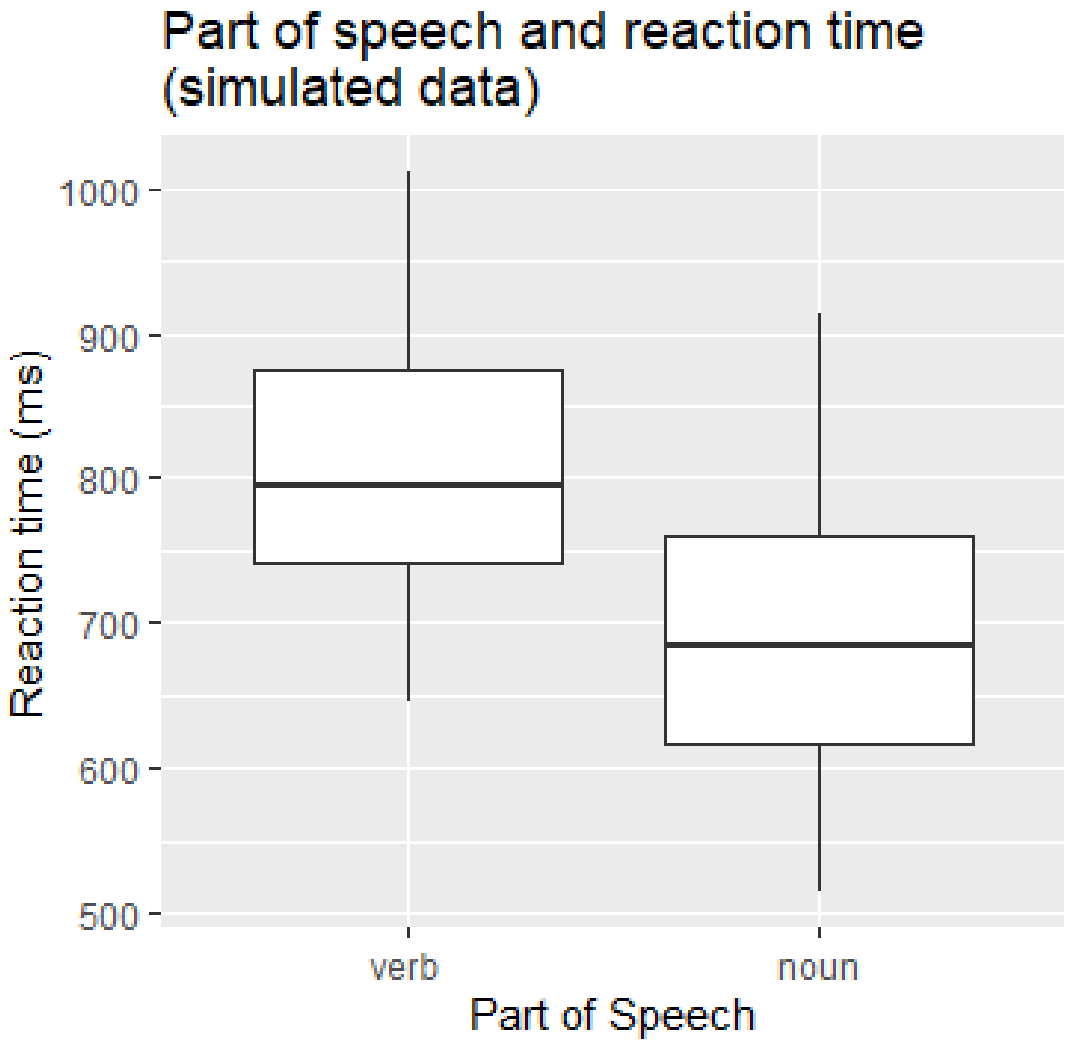
$1/y = a + bx$

Categorical Variables

Regressions can also use categorical variables. Let's say we have the following equation:

RT ~ POS
POS: {noun, verb}

How does the computer make the regression if one of the variables has no numbers?



Categorical Variables

```
> fposM = lm(rt  ~ pos, data = fpos)
> summary(fposM)

Call:
lm(formula = rt ~ pos, data = fpos)

Residuals:
    Min      1Q  Median      3Q     Max 
-175.82  -69.13  -12.52   72.53  222.74 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   690.39      13.22  52.204  < 2e-16 ***
posverb       116.22      18.70   6.214 1.26e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93.51 on 98 degrees of freedom
Multiple R-squared:  0.2826,	Adjusted R-squared:  0.2753 
F-statistic: 38.61 on 1 and 98 DF,  p-value: 1.26e-08
```

Categorical Variables

<u>Original equation</u>

`rt ~ β*pos`

(pos is either verb or noun)

<u>New equation</u>

`rt ~ α + β*posverb`

If we want to know the value of RT for `posnoun`, then we make `posverb` = 0
If we want to know the value of RT for `posverb`, then we make `posverb` = 1

We call these **dummy variables**.
The first category, alphabetically, is the **reference level**.

Categorical Variables

```
> fposM = lm(rt  ~ pos, data = fpos)
> summary(fposM)

Call:
lm(formula = rt ~ pos, data = fpos)

Residuals:
    Min      1Q  Median      3Q     Max 
-175.82  -69.13  -12.52   72.53  222.74 

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---
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Residual standard error: 93.51 on 98 degrees of freedom
Multiple R-squared:  0.2826,	Adjusted R-squared:  0.2753 
F-statistic: 38.61 on 1 and 98 DF,  p-value: 1.26e-08
```

The computer created the variable:

posverb

Alphabetically, posnoun comes first, so this will be our REFERENCE LEVEL.

Categorical Variables

```
> fposM = lm(rt  ~ pos, data = fpos)
> summary(fposM)

Call:
lm(formula = rt ~ pos, data = fpos)

Residuals:
    Min      1Q  Median      3Q     Max 
-175.82  -69.13  -12.52   72.53  222.74 

Coefficients:
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(Intercept)   690.39      13.22  52.204  < 2e-16 ***
posverb       116.22      18.70   6.214 1.26e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93.51 on 98 degrees of freedom
Multiple R-squared:  0.2826,	Adjusted R-squared:  0.2753 
F-statistic: 38.61 on 1 and 98 DF,  p-value: 1.26e-08
```

posnoun is being compared to posverb (the reference level).

Here, the RT for posnoun is significantly different from the RT for posverb (β=116, t(98)=6.2, p<0.0001, R^2=0.28).

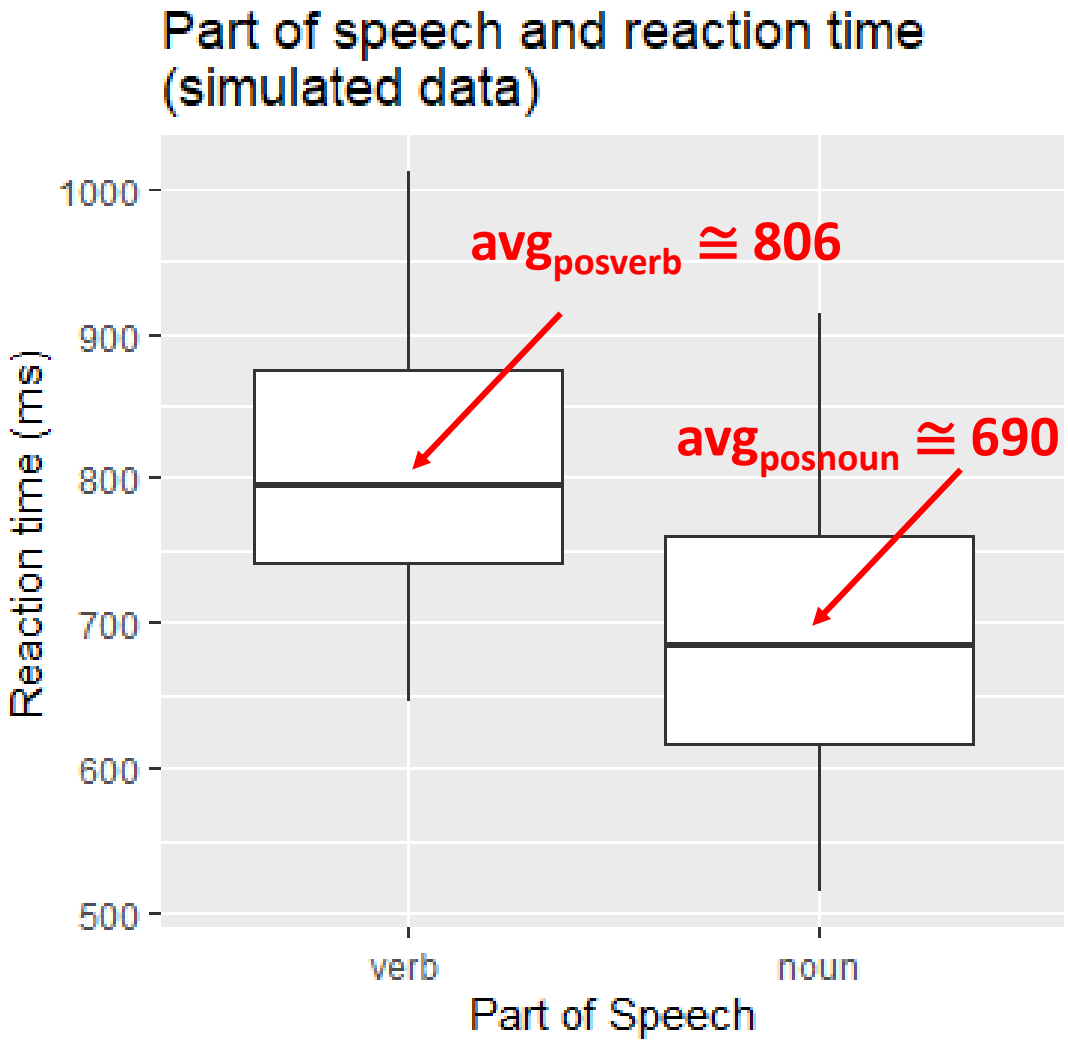
Categorical Variables

What does the coefficient/beta mean?

$\beta_{posverb} = 116$
Intercept = 690

Here, the intercept is the average of the reference level (posnoun).

The average of posverb is:
Intercept + $\beta_{posverb}$ =
116+690 = 806



Categorical Variables

```
> fposM = lm(rt  ~ pos, data = fpos)
> summary(fposM)
Call:
lm(formula = rt ~ pos, data = fpos)
Residuals:
    Min      1Q  Median      3Q     Max
-175.82  -69.13  -12.52   72.53  222.74
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   690.39      13.22  52.204  < 2e-16 ***
posverb       116.22      18.70   6.214 1.26e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 93.51 on 98 degrees of freedom
Multiple R-squared:  0.2826,  Adjusted R-squared:  0.2753
F-statistic: 38.61 on 1 and 98 DF,  p-value: 1.26e-08
```

A linear regression was run on 100 simulated words, with the part-of-speech as a categorical independent variable (levels: noun, verb) and the reaction time in milliseconds as the dependent variable. Nouns have a significantly shorter RT (mean=690ms, SE=13ms) than verbs (mean=806ms, β=116ms, SE=19ms, t(98)=-6.2, p<0.0001, R^2=0.28).

Categorical Variables

```
> fpos$pos <- relevel(fpos$pos, ref="verb")
> fposM = lm(rt  ~ pos, data = fpos)
> summary(fposM)

Call:
lm(formula = rt ~ pos, data = fpos)

Residuals:
    Min      1Q  Median      3Q     Max 
-175.82  -69.13  -12.52   72.53  222.74 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   806.60      13.22  60.991  < 2e-16 ***
posnoun      -116.22      18.70  -6.214 1.26e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93.51 on 98 degrees of freedom
Multiple R-squared:  0.2826,	Adjusted R-squared:  0.2753 
F-statistic: 38.61 on 1 and 98 DF,  p-value: 1.26e-08
```

It's very easy to change the reference level.

Now, posverb is the reference level. The intercept is the average for `posverb` = 806ms.

The average for `posnoun` is 806-116 = 690ms.

Practice: Categorical Variables

German and Dutch exist in a *Germanic Continuum* of very closely related dialects.

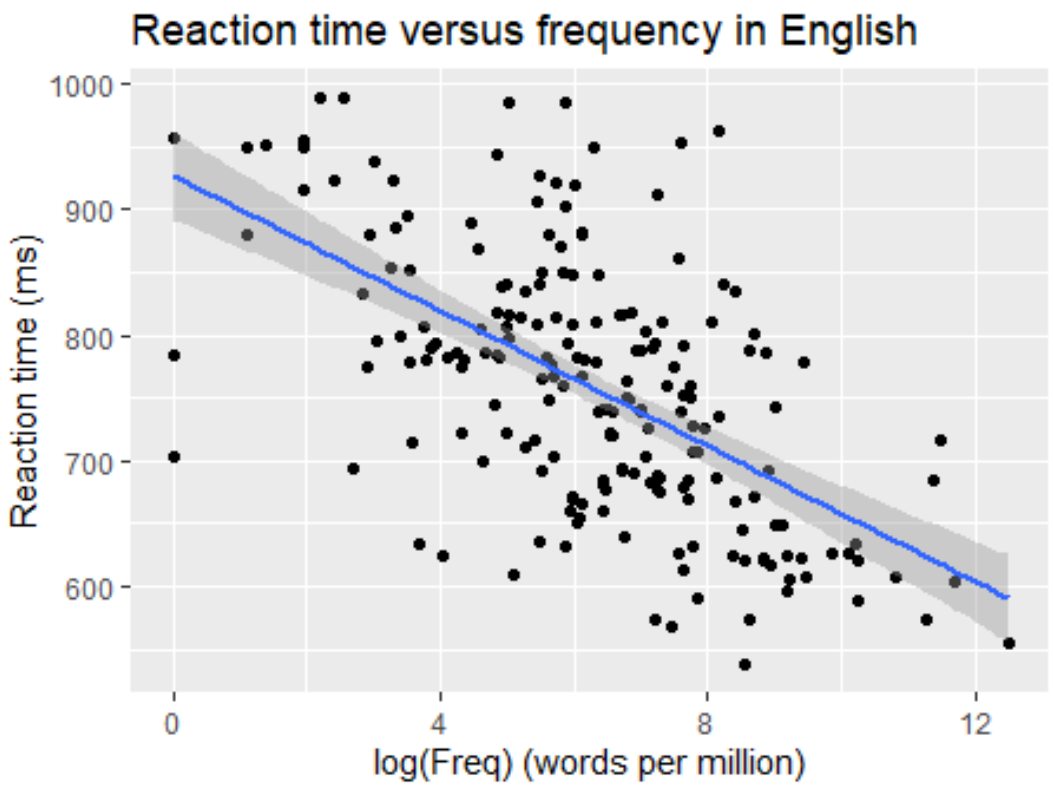
Dutch:	Ik denk dat ik **kan gaan**.	Modal+Verb
	I think that I can go.	
German:	Ich denke daß ich **gehen kann**.	Verb+Modal
	I think that I go can.	

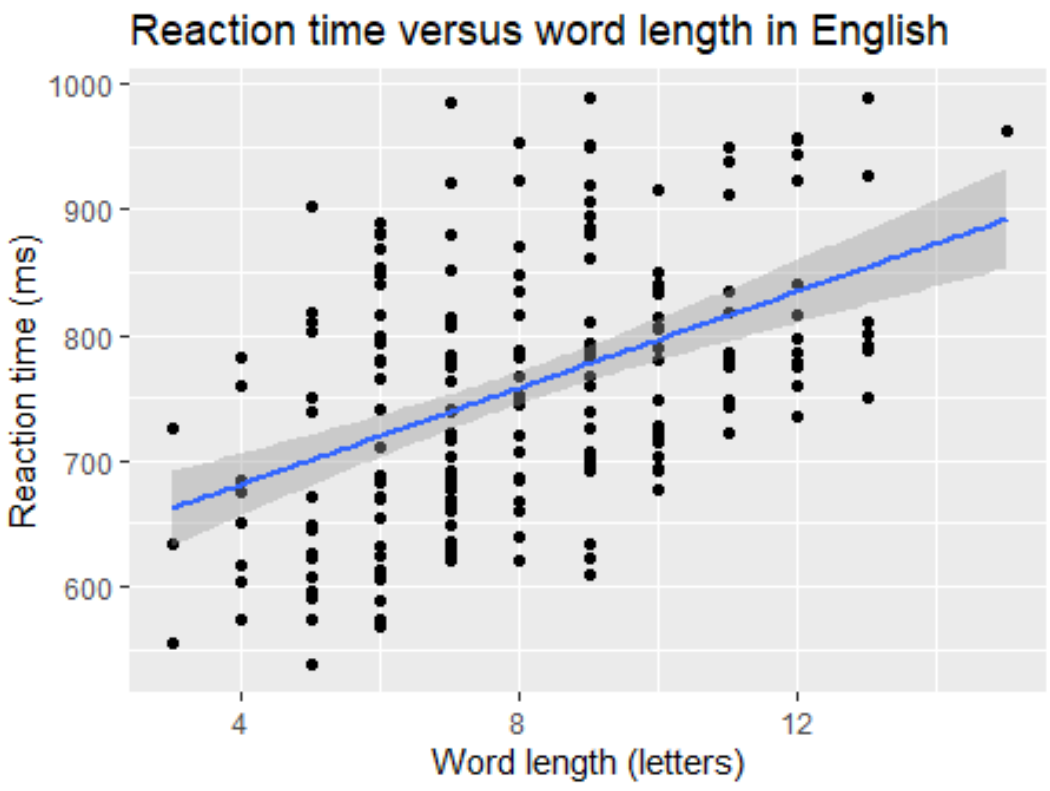
The dataset `german-modals-20.csv` has information on the percentage of sentences where the modal comes before the verb (as in Dutch). It contains data for 3 nearby villages in Belgium, Germany and the Netherlands. 20 people were interviewed in each village.

Is the percentage of Modal+Verb different depending on the country where the village is?

Multiple variables

Many phenomena are related to more than one variable. For example, RT in English can be related to the frequency of words, as well as the length of the words.





Multiple variables

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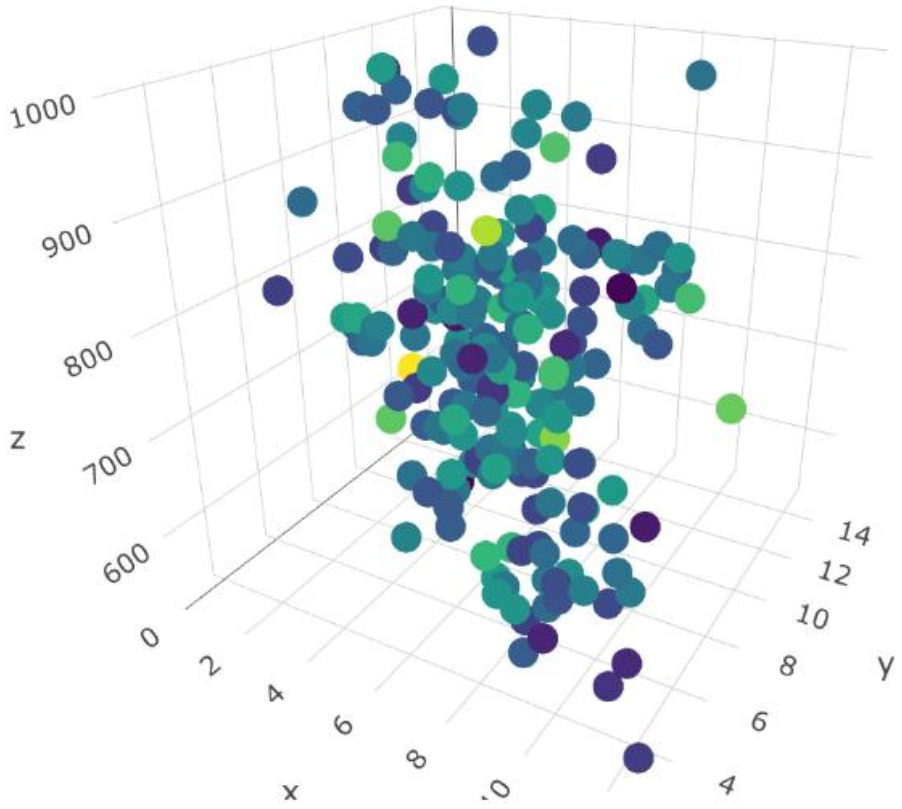
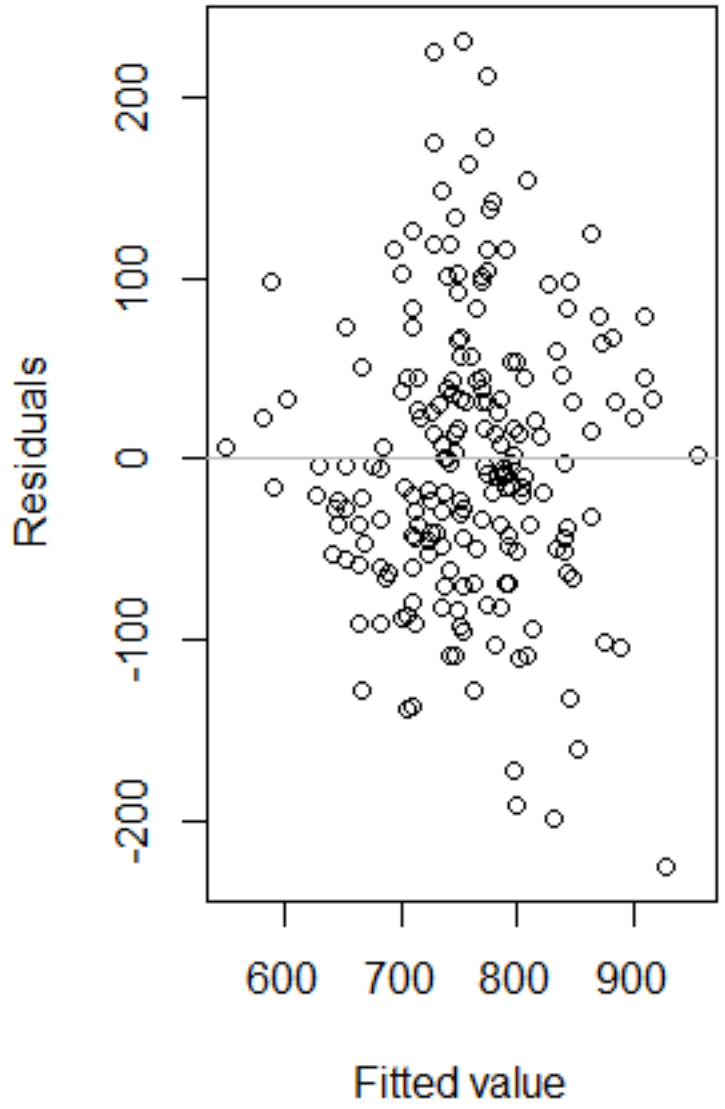
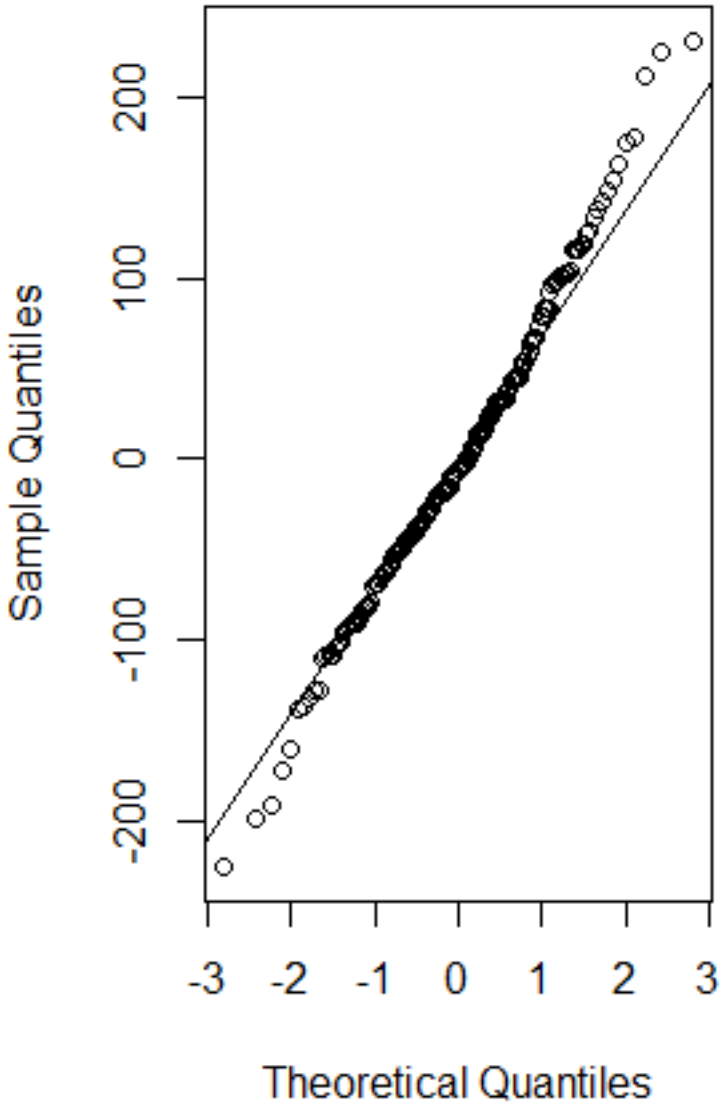


Multiple variables





Multiple variables

```
> # Shapiro-Wilk Test of normality for the residuals
> shapiro.test(elpMultModel$residuals)

        Shapiro-Wilk normality test

data:  elpMultModel$residuals
W = 0.99071, p-value = 0.2261

> # Non-constant variance test
> ncvTest(elpMultModel)
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 3.936709, Df = 1, p = 0.047243

> # Autocorrelation
> durbinWatsonTest(elpMultModel)
 lag Autocorrelation D-W Statistic p-value
   1     -0.06028407      2.107324   0.472
 Alternative hypothesis: rho != 0
```

Multiple variables

```
> elpMultModel = lm(I_Mean_RT  ~ Log_Freq_HAL + Length, data = elpM)
> summary(elpMultModel)
Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL + Length, data = elpM)

Residuals:
     Min       1Q   Median       3Q      Max 
-224.841  -48.621   -5.095   44.817  230.996 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   794.916     28.504  27.888  < 2e-16 ***
Log_Freq_HAL  -22.911      2.562  -8.944 2.76e-16 ***
Length         13.374      2.362   5.662 5.23e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 78.62 on 197 degrees of freedom
Multiple R-squared:  0.4343,	Adjusted R-squared:  0.4285 
F-statistic: 75.62 on 2 and 197 DF,  p-value: < 2.2e-16
```

Multiple variables

```
> elpMultModel = lm(I_Mean_RT ~ Log_Freq_HAL + Length, data = elpM)
> summary(elpMultModel)
Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL + Length, data = elpM)

Residuals:
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Multiple R-squared:  0.4343,	Adjusted R-squared:  0.4285 
F-statistic: 75.62 on 2 and 197 DF,  p-value: < 2.2e-16
```

We add variables

Multiple variables

```
> elpMultModel = lm(I_Mean_RT  ~ Log_Freq_HAL + Length, data = elpM)
> summary(elpMultModel)
Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL + Length, data = elpM)

Residuals:
     Min       1Q   Median       3Q      Max 
-224.841  -48.621   -5.095   44.817  230.996 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   794.916     28.504  27.888  < 2e-16 ***
Log_Freq_HAL  -22.911      2.562  -8.944 2.76e-16 ***
Length         13.374      2.362   5.662 5.23e-08 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 78.62 on 197 degrees of freedom
Multiple R-squared:  0.4343,	Adjusted R-squared:  0.4285 
F-statistic: 75.62 on 2 and 197 DF,  p-value: < 2.2e-16
```

We have an overall ANOVA, that tell us that something is significant.

Multiple variables

```
> elpMultModel = lm(I_Mean_RT  ~ Log_Freq_HAL + Length, data = elpM)
> summary(elpMultModel)
Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL + Length, data = elpM)

Residuals:
     Min       1Q   Median       3Q      Max 
-224.841  -48.621   -5.095   44.817  230.996 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   794.916     28.504  27.888  < 2e-16 ***
Log_Freq_HAL  -22.911      2.562  -8.944 2.76e-16 ***
Length         13.374      2.362   5.662 5.23e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 78.62 on 197 degrees of freedom
Multiple R-squared:  0.4343,	Adjusted R-squared:  0.4285 
F-statistic: 75.62 on 2 and 197 DF,  p-value: < 2.2e-16
```

Each variable has its own coefficient and its own t-test, to verify that the coefficient is different from zero.

Multiple variables

```
> elpMultModel = lm(I_Mean_RT  ~ Log_Freq_HAL + Length, data = elpM)
> summary(elpMultModel)
Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL + Length, data = elpM)
Residuals:
     Min       1Q   Median       3Q      Max
-224.841  -48.621   -5.095   44.817  230.996
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   794.916     28.504  27.888  < 2e-16 ***
Log_Freq_HAL  -22.911      2.562  -8.944 2.76e-16 ***
Length         13.374      2.362   5.662 5.23e-08 ***
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Residual standard error: 78.62 on 197 degrees of freedom
Multiple R-squared:  0.4343,  Adjusted R-squared:  0.4285
F-statistic: 75.62 on 2 and 197 DF,  p-value: < 2.2e-16
```

Equation:

$$RT = 794.9 + (-22.9)*Log_Freq_HAL + (13.4)*Length + \varepsilon$$

Multiple variables

```
> elpMultModel = lm(I_Mean_RT  ~ Log_Freq_HAL + Length, data = elpM)
> summary(elpMultModel)
Call:
lm(formula = I_Mean_RT ~ Log_Freq_HAL + Length, data = elpM)
Residuals:
     Min       1Q   Median       3Q      Max
-224.841  -48.621   -5.095   44.817  230.996
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   794.916     28.504  27.888  < 2e-16 ***
Log_Freq_HAL  -22.911      2.562  -8.944 2.76e-16 ***
Length         13.374      2.362   5.662 5.23e-08 ***
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Residual standard error: 78.62 on 197 degrees of freedom
Multiple R-squared:  0.4343,  Adjusted R-squared:  0.4285
F-statistic: 75.62 on 2 and 197 DF,  p-value: < 2.2e-16
```

A linear regression was run on 200 English words, with log(Frequency) and word length as independent variables, and the reaction time in milliseconds as the dependent variable. Longer words have significantly longer RTs (β=13.4, SE=2.4, t(198)=5.7, p<0.0001), while more frequent words have significantly shorter RTs (β=-22.9, SE=2.6, t(198)=-8.9, p<0.0001). The effect of the two variables is R^2=0.43.

A New Assumption: Collinearity

Collinearity is a situation where two predictor variables are closely related. It is a problem because the effects of one are confused with the effects of the other.

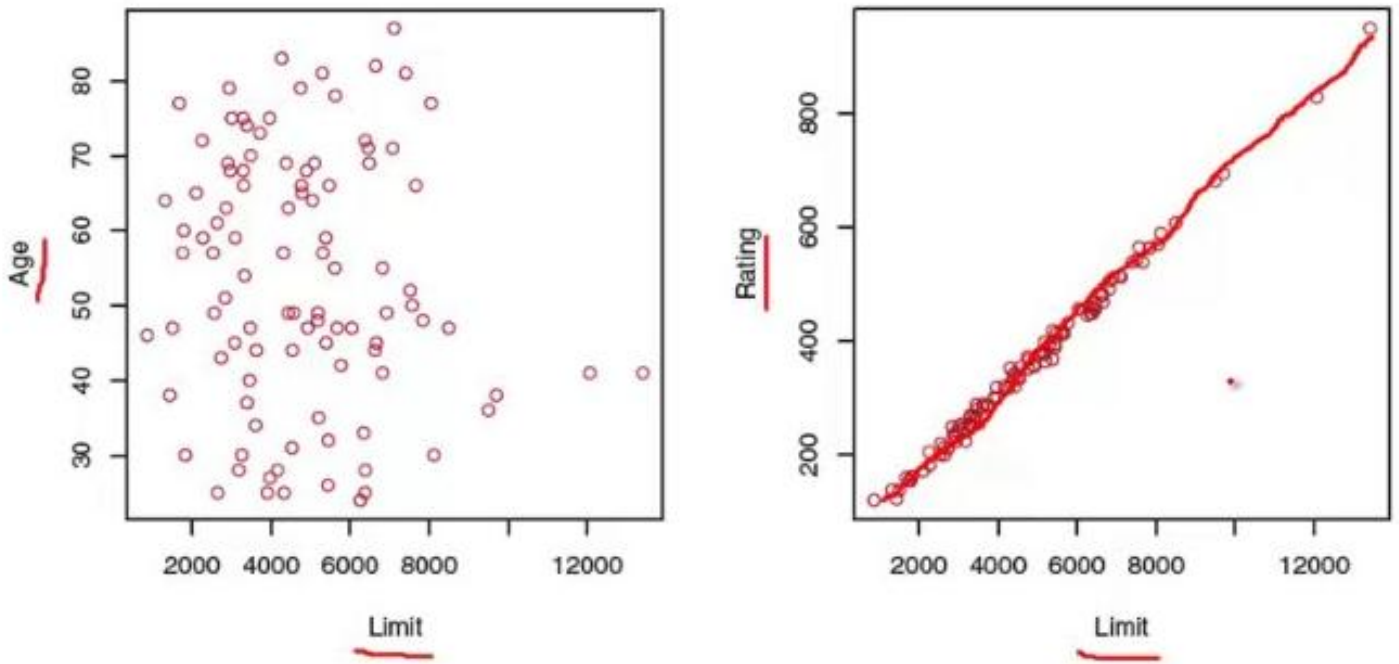
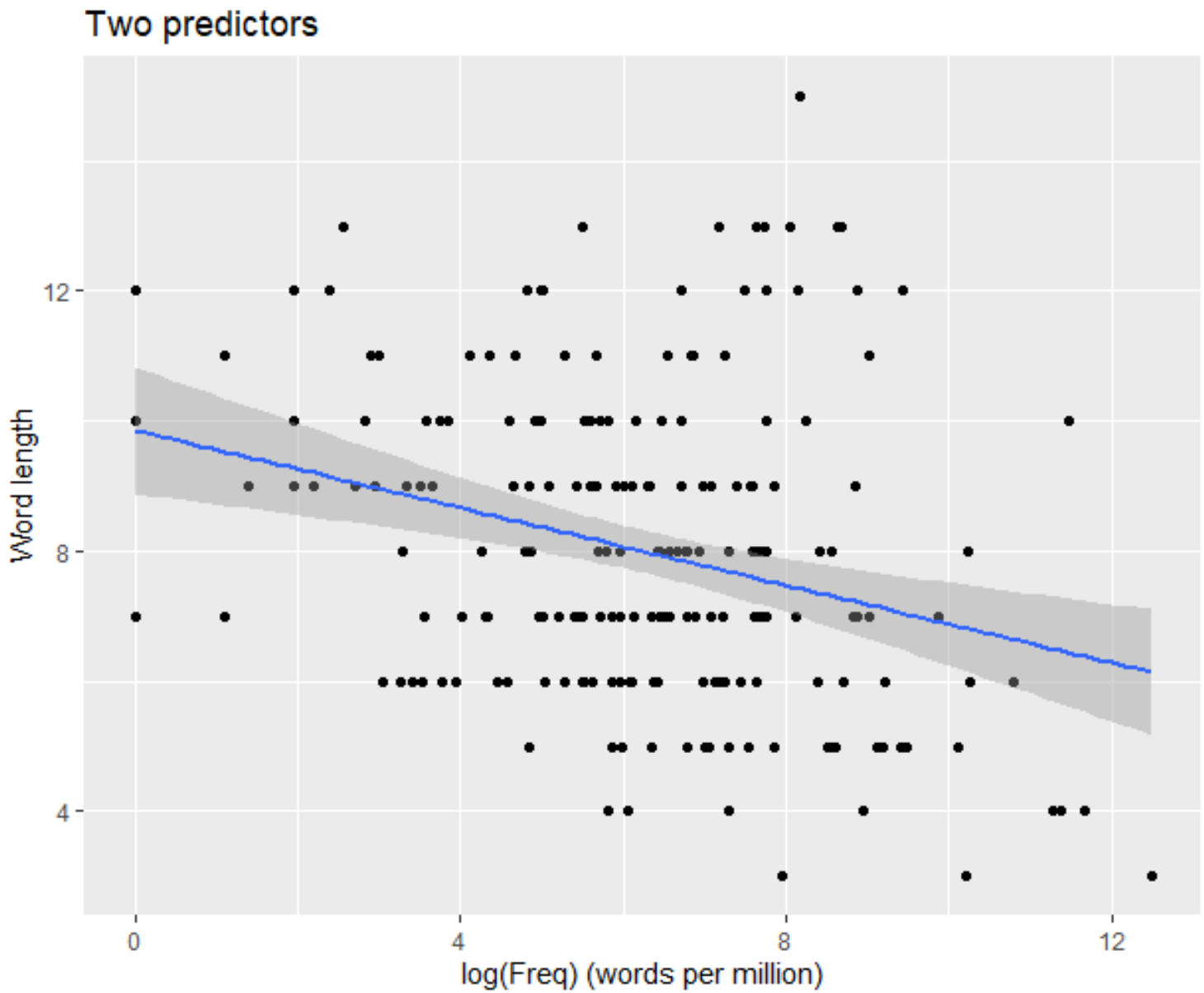


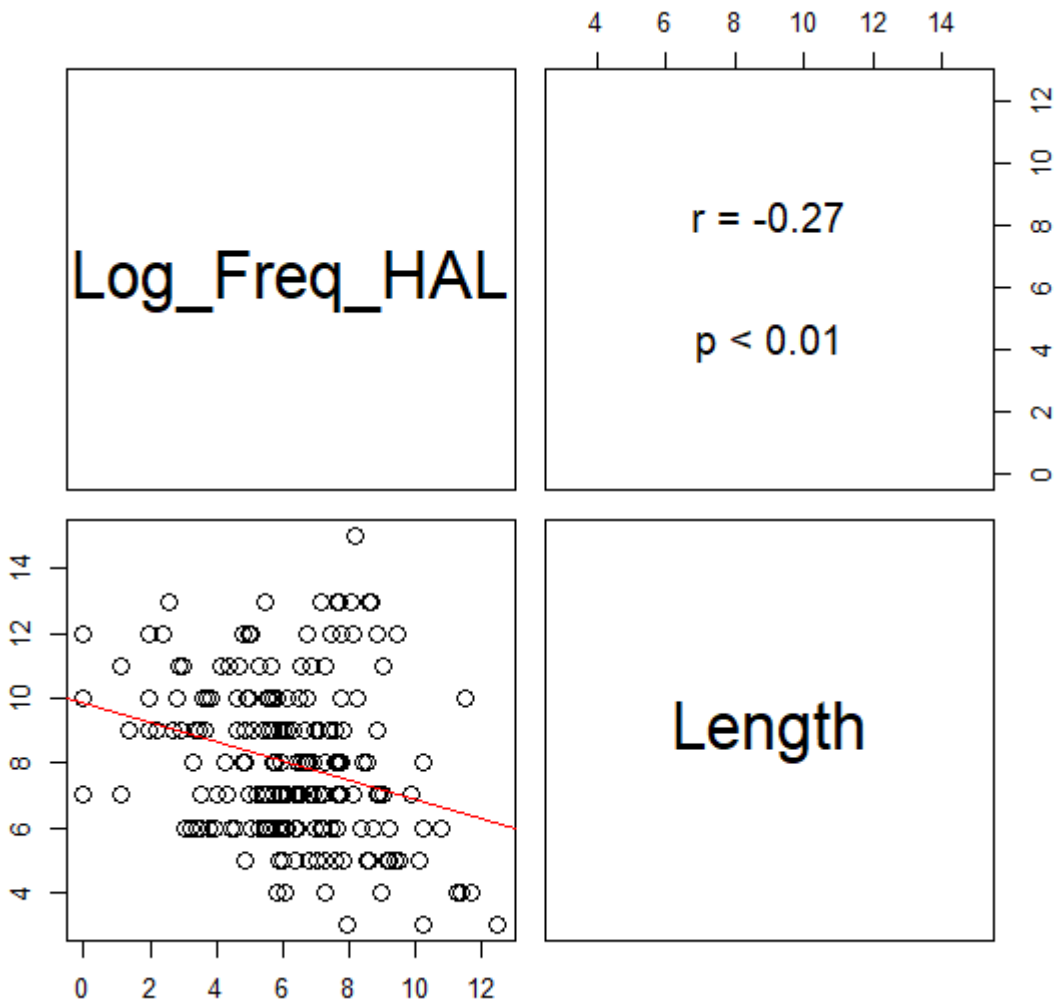
FIGURE 3.14. *Scatterplots of the observations from the* `Credit` *data set.* Left: *A plot of* `age` *versus* `limit`. *These two variables are not collinear.* Right: *A plot of* `rating` *versus* `limit`. *There is high collinearity.*

A New Assumption: Collinearity



A New Assumption: Collinearity

```
pairs(~ Log_Freq_HAL + Length, data=elpM, cex.labels=2.5, cex=1.5,
lower.panel=panel.lm, upper.panel=panel.cor.res)
```



A New Assumption: Collinearity

The **variance inflation factor** is a formal measure of collinearity. If the VIF of each predictor variable is larger than the VIF of the model, then that variable might be strongly multicollinear with another predictor variable.

If the VIFj is greater than 10, it is very likely multicollinear with another predictor variable.

$$VIF_j = \frac{1}{1 - R_j^2} \qquad VIF_{model} = \frac{1}{1 - R_{model}^2}$$

A New Assumption: Collinearity

Fun fact: We are willing to tolerate **A LOT** of correlation before we declare "collinearity!". For the VIF_j to be greater than 10, the r needs to be 0.949!

$$VIF_j = \frac{1}{1 - R_j^2} \Rightarrow$$

$$10(1 - R^2) = 1 \Rightarrow$$

$$R^2 = 1 - \frac{1}{10} \Rightarrow$$

$$R^2 = 0.9 \Rightarrow$$

$$r = 0.949$$

A New Assumption: Collinearity

```
> vif(elpMultModel)
Log_Freq_HAL       Length
    1.081441     1.081441
```

Practice: Multiple variables

Exercise: RT in English

(Assume that the variance is constant)